

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	4
Nozzle Schedule:	6
Bill of Material:	8
MDMT Summary:	9
Warnings and Errors:	11
Input Echo:	12
XY Coordinate Calculations:	24
Internal Pressure Calculations:	25
Element: Left Head	26
Element: Shell-1	28
Element: Shell-2	30
Element: Shell-3	32
Element: Right Head	34
Element: Sump Head	36
Shop Pressure Test Results:	38
Field Pressure Test Results:	40
External Pressure Calculations:	42
Element and Detail Weights:	43
Nozzle Flange MAWP:	46
Wind Load Calculation:	48
Earthquake Load Calculation:	50
Center of Gravity Calculation:	52
Saddle Calcs: Operating Case:	53
Saddle Calcs: Test Case:	68
Saddle Calcs: Ext. Pressure:	81
Nozzle Calcs.: MH-1_24"	93
Nozzle Calcs.: N1_8"	97
Nozzle Calcs.: N4_2"	107
Nozzle Calcs.: K3A_3"	115
Nozzle Calcs.: K3B_3"	121
Nozzle Calcs.: K3C_3"	127
Nozzle Calcs.: K3D_3"	133
Nozzle Calcs.: K3E_3"	139
Nozzle Calcs.: K3F_3"	145
Nozzle Calcs.: K6_1"	151
Nozzle Calcs.: N5_2"	154
Nozzle Calcs.: K1A_3"	162
Nozzle Calcs.: K1B_3"	168
Nozzle Calcs.: K2A_1.5"	174
Nozzle Calcs.: K2B_1.5"	177
Nozzle Calcs.: K5_1.5"	180
Nozzle Calcs.: K4_1.5"	183
Nozzle Calcs.: N3_6"	186
Nozzle Calcs.: K7_8"	196
Nozzle Calcs.: Boot	206
Nozzle Calcs.: MH2_24"	210
Nozzle Calcs.: N2_2"	214



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\V-310027%.pvdb

Date of Analysis : May 20, 2026 1:47pm

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1700.000 mm ID	
Vessel Design Length, Tangent to Tangent	7950.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	80 °C
Internal Design Pressure	2.600 MPa
External Design Temperature	80 °C
External Design Pressure	0.103 MPa
Maximum Allowable Working Pressure	2.62 MPa
External Max. Allowable Working Pressure	0.3 MPa
Shop Test Pressure	3.403 MPa
Field Test Pressure	3.403 MPa
Required Minimum Design Metal Temperature	-33.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Left Head	2.611	0.10	2.660	1.5000	No	No
Shell-1	2.611	0.10	2.630	1.5000	N/A	No
Shell-2	2.611	0.10	2.630	1.5000	N/A	No
Shell-3	2.611	0.10	2.630	1.5000	N/A	No
Right Head	2.611	0.10	2.660	1.5000	No	No
Sump Head	2.600	0.10	9.620	1.5000	No	No

Liquid Level: 1700.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.631

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	22.0	18.0	17.6	6.0	1.00	1.00
Cylinder	3050.0	3000.0	18.0	18.0	17.8	12.2	1.00	1.00
Cylinder	6050.0	3000.0	18.0	18.0	17.8	12.2	1.00	1.00



Cylinder	7900.0	1850.0	18.0	18.0	17.8	12.2	1.00	1.00
Ellipse	7950.0	50.0	22.0	18.0	17.6	6.0	1.00	1.00
Ellipse	7630.0	0.0	16.0	12.0	4.3	2.3	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1_8"	WRC-297	0.978	Shell	Passed
N4_2"	WRC-297	0.986	Shell	Passed
N5_2"	WRC-297	0.986	Shell	Passed
N3_6"	WRC-297	0.986	Shell	Passed
K7_8"	WRC-297	0.978	Shell	Passed
N2_2"	WRC-107/537	0.244	n/a	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	125.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1180.000	mm
Wear Pad Width	180.000	mm
Wear Pad Thickness	6.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	1550.000	mm
Baseplate Length	1575.000	mm
Baseplate Thickness	16.000	mm
Baseplate Width	150.000	mm
Number of Ribs (including outside ribs)	3	
Rib Thickness	10.000	mm
Web Thickness	10.000	mm
Height of Center Web	290.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	232457.70	N
Maximum Transverse Saddle Shear Load	29143.23	N
Maximum Longitudinal Saddle Shear Load	25766.92	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	215536.84	N
Maximum Transverse Saddle Shear Load	9617.27	N
Maximum Longitudinal Saddle Shear Load	4452.90	N

Weights:

Fabricated - Bare w/o removable internals	10822.4	kgm
Shop Test - Fabricated + water (full)	33162.3	kgm
Shipping - Fab. + rem. intls.+ shipping app.	13707.6	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	16717.6	kgm
Empty - Fab. + intls. + details + wghts.	16717.6	kgm
Operating - Empty + operating liquid (no ca)	28908.3	kgm
Field Test - Empty weight + test medium (full)	36047.4	kgm

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
MH-1_24"	2.62	OK	...	...		OK	Passed
N1_8"	2.63	OK	...	OK	8.66	OK	Passed
N4_2"	2.63	...	...	OK	6.30	OK	No Calc[*]
K3A_3"	2.63	OK	...	OK	7.23	OK	Passed
K3B_3"	2.63	OK	...	OK	7.23	OK	Passed
K3C_3"	2.63	OK	...	OK	7.23	OK	Passed
K3D_3"	2.63	OK	...	OK	7.23	OK	Passed
K3E_3"	2.63	OK	...	OK	7.23	OK	Passed
K3F_3"	2.63	OK	...	OK	7.23	OK	Passed
K6_1"	2.63	...	...	OK	4.92	OK	No Calc[*]
N5_2"	2.63	...	...	OK	6.30	OK	No Calc[*]
K1A_3"	2.63	OK	...	OK	7.23	OK	Passed
K1B_3"	2.63	OK	...	OK	7.23	OK	Passed
K2A_1.5"	2.63	...	...	OK	6.02	OK	No Calc[*]
K2B_1.5"	2.63	...	...	OK	6.02	OK	No Calc[*]
K5_1.5"	2.63	...	...	OK	6.02	OK	No Calc[*]
K4_1.5"	2.63	...	...	OK	6.02	OK	No Calc[*]
N3_6"	2.63	OK	...	OK	7.72	OK	Passed
K7_8"	2.63	OK	...	OK	8.66	OK	Passed
Boot	2.63	OK	...	OK	9.83	OK	Passed
MH2_24"	2.62	OK	...	...		OK	Passed
N2_2"	9.62	...	...	OK	4.32	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2.62 Nozzle : MH224"
 Minimum MAWP Shells/Flanges : 2.62 Element : Shell-3
 Minimum MAPnc Shells/Flanges : 2.89 Element : Shell-3

Computed Vessel M.A.W.P. : 2.62 MPa

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	MH-1_24"	0.000	0.000	1225.200
20	N1_8"	310.000	0.000	393.350
20	N4_2"	200.000	180.000	117.328
20	K3A_3"	2000.000	54.828	158.400
20	K3B_3"	2000.000	125.172	158.400
20	K3C_3"	2300.000	54.828	158.400
20	K3D_3"	2300.000	125.172	158.400
20	K3E_3"	2600.000	54.828	158.400
20	K3F_3"	2600.000	125.172	158.400
20	K6_1"	580.000	0.000	86.848
20	N5_2"	500.000	180.000	117.328
30	K1A_3"	4700.000	54.828	158.400
30	K1B_3"	4700.000	125.172	158.400
30	K2A_1.5"	4500.000	54.828	103.104
30	K2B_1.5"	4500.000	125.172	103.104
30	K5_1.5"	5100.000	103.322	103.104
30	K4_1.5"	4900.000	103.322	103.104
40	N3_6"	7690.000	180.000	298.659
40	K7_8"	7000.000	0.000	393.350
40	Boot	7630.000	0.000	615.600
50	MH2_24"	0.000	0.000	1225.200



60	N2_2"	0.000	0.000	107.600
----	-------	-------	-------	---------

The nozzle spacing is computed by the following:

$= \sqrt{ll^2 + lc^2}$ where

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K6_1"	1.000 in	Actual	LWN	2.120	14.224	...	...	444.68	300
K2A_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	257.02	300
K2B_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	257.02	300
K5_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	235.75	300
K4_1.5"	1.500 in	Actual	LWN	2.760	16.002	...	...	235.75	300
N4_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	260.47	600
N5_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	255.65	600
N2_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	256.62	600
K3A_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	268.94	300
K3B_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	268.94	300
K3C_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	268.94	300
K3D_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	268.94	300
K3E_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	268.94	300
K3F_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	268.94	300
K1A_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	268.94	300
K1B_3"	3.000 in	Actual	LWN	4.620	20.574	...	...	268.94	300
N3_6"	6.000 in	80	WNF	6.625	10.973	...	...	192.54	300
N1_8"	8.000 in	80	WNF	8.625	12.700	...	...	363.76	300
K7_8"	8.000 in	80	WNF	8.625	12.700	...	...	363.76	300
Boot	12.000 in	Actual	None	12.945	12.000	...	...	547.61	...
MH-1_24"	24.000 in	Actual	WNF	24.945	12.000	...	...	190.37	300
MH2_24"	24.000 in	Actual	WNF	24.945	12.000	...	...	190.37	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K6_1"	SA-350 LF2	18.000	10.000	...	...	...
K2A_1.5"	SA-350 LF2	18.000	10.000	...	...	...
K2B_1.5"	SA-350 LF2	18.000	10.000	...	...	...
K5_1.5"	SA-350 LF2	18.000	10.000	...	...	...
K4_1.5"	SA-350 LF2	18.000	10.000	...	...	...
N4_2"	SA-350 LF2	18.000	10.000	...	...	...
N5_2"	SA-350 LF2	18.000	10.000	...	...	...
N2_2"	SA-350 LF2	12.000	10.000	...	...	...
K3A_3"	SA-350 LF2	18.000	10.000	...	...	...
K3B_3"	SA-350 LF2	18.000	10.000	...	...	...
K3C_3"	SA-350 LF2	18.000	10.000	...	...	...
K3D_3"	SA-350 LF2	18.000	10.000	...	...	...
K3E_3"	SA-350 LF2	18.000	10.000	...	...	...
K3F_3"	SA-350 LF2	18.000	10.000	...	...	...
K1A_3"	SA-350 LF2	18.000	10.000	...	...	...
K1B_3"	SA-350 LF2	18.000	10.000	...	...	...
N3_6"	SA-350 LF2	18.000	10.000	...	...	...
N1_8"	SA-350 LF2	18.000	10.000	...	...	...
K7_8"	SA-350 LF2	18.000	10.000	...	...	...
Boot	SA-350 LF2	18.000	10.000	...	...	...
MH-1_24"	SA-350 LF2	18.000	10.000	...	...	...
MH2_24"	SA-350 LF2	18.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.



Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K6_1"	580.000	0.0	426.25	0.00	Shell-1
K2A_1.5"	4500.000	54.8	207.96	0.00	Shell-2
K2B_1.5"	4500.000	125.2	207.96	0.00	Shell-2
K5_1.5"	5100.000	103.3	207.96	0.00	Shell-2
K4_1.5"	4900.000	103.3	207.96	0.00	Shell-2
N4_2"	200.000	180.0	241.43	0.00	Shell-1
N5_2"	500.000	180.0	236.60	0.00	Shell-1
N2_2"	...	0.0	241.43	0.00	Sump Head
K3A_3"	2000.000	54.8	200.02	0.00	Shell-1
K3B_3"	2000.000	125.2	200.02	0.00	Shell-1
K3C_3"	2300.000	54.8	200.02	0.00	Shell-1
K3D_3"	2300.000	125.2	200.02	0.00	Shell-1
K3E_3"	2600.000	54.8	200.02	0.00	Shell-1
K3F_3"	2600.000	125.2	200.02	0.00	Shell-1
K1A_3"	4700.000	54.8	200.02	0.00	Shell-2
K1B_3"	4700.000	125.2	200.02	0.00	Shell-2
N3_6"	7690.000	180.0	164.97	0.00	Shell-3
N1_8"	310.000	0.0	332.27	0.00	Shell-1
K7_8"	7000.000	0.0	332.27	0.00	Shell-3
Boot	7630.000	0.0	500.00	0.00	Shell-3
MH-1_24"	...	0.0	118.38	0.00	Left Head
MH2_24"	...	0.0	118.38	0.00	Right Head



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 22.0mm THK X 1700.0mm ID X 50.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm THK X 300.0mm ID X 50.0mm	SA-516 70
2	INSULATION: 475mm X 90mm THK	
2	SADDLE: 125.0mm X 120 DEG	SA-516 70
1	PLATFORM: 1200mm X 1736mm WIDE	
1	LADDER: 4000.0mm	...
2	INSULATION: 3000mm X 90mm THK	
1	INSULATION: 1850mm X 90mm THK	
1	INSULATION: 125mm X 90mm THK	
2	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-350 LF2
2	CYLINDRICAL SEGMENT 18.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 18.0mm THK X 2292.7mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 3048.0mm X 1850.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 2292.7mm X 1850.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Left Head	[10]	B	-11	-11	-29	0.982	18.000	1.00	No
Left Head	[7]	B	-5	-16	-29	0.800	22.000	1.00	No
Shell-1	[8]	B	-11	-11	-29	0.994	18.000	1.00	No
Shell-2	[8]	B	-11	-11	-29	0.994	18.000	1.00	No
Shell-3	[8]	B	-11	-11	-29	0.994	18.000	1.00	No
Right Head	[10]	B	-11	-11	-29	0.982	18.000	1.00	No
Right Head	[7]	B	-5	-16	-29	0.800	22.000	1.00	No
Sump Head	[10]	B	-23	-104	-29	0.271	12.000	1.00	No
Sump Head	[7]	B	-14	-104	-29	0.200	16.000	1.00	No
Left Saddle	[24]	B	-29	-29	-29	0.994	6.000	1.00	No
Right Saddle	[24]	B	-29	-29	-29	0.994	6.000	1.00	No
MH-124"	[1]	B	-11	-29	-29	0.877	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-77					
N18"	[1]	B	-11	-29	-29	0.983	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-77					
N42"	[1]	B	-13	-104	-29	0.034	16.764	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3A3"	[1]	B	-11	-29	-29	0.984	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3B3"	[1]	B	-11	-29	-29	0.987	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3C3"	[1]	B	-11	-29	-29	0.984	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3D3"	[1]	B	-11	-29	-29	0.987	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3E3"	[1]	B	-11	-29	-29	0.984	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K3F3"	[1]	B	-11	-29	-29	0.987	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K61"	[1]	B	-18	-104	-29	0.021	14.224	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N52"	[1]	B	-13	-104	-29	0.034	16.764	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K1A3"	[1]	B	-11	-29	-29	0.984	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K1B3"	[1]	B	-11	-29	-29	0.987	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K2A1.5"	[1]	B	-14	-104	-29	0.027	16.002	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K2B1.5"	[1]	B	-14	-104	-29	0.027	16.002	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K51.5"	[1]	B	-14	-104	-29	0.027	16.002	1.00	No
Nozzle Flg	[5]	!	-46	-46					
K41.5"	[1]	B	-14	-104	-29	0.027	16.002	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N36"	[1]	B	-11	-29	-29	0.988	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-76					
K78"	[1]	B	-11	-29	-29	0.983	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-77					
Boot	[1]	B	-11	-29	-29	0.983	18.000	1.00	No
MH224"	[1]	B	-11	-29	-29	0.877	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-77					
N22"	[1]	B	-23	-104	-29	0.243	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					

Warmest MDMT: -5 -11

Required Minimum Design Metal Temperature -33.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:



- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	2.6002	MPa
Design Internal Temperature	80.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-33.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000



Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117



0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Left Head
Distance "FROM" to "TO"	50 mm
Inside Diameter	1700 mm
Element Thickness	18 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.6002 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	243 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0



Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1700 mm
Liquid Density	0.0006299 kg/cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-425 mm
Height/Length of Insulation	475 mm
Thickness of Insulation	90 mm
Density	0.40000E-04 kg/cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	MH-124"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	7833 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT
Element From Node	10
Detail Type	Weight
Detail ID	Manway Davit
Dist. from "FROM" Node / Offset dist	-425 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	0 mm

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	1700 mm
Element Thickness	18 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.6002 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1550 mm
Width of Saddle	125 mm
Height of Saddle at Bottom	1180 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	180 mm
Thickness of Wear Plate	6 mm



Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	20	
Detail Type	Platform	
Detail ID	PLATFORM	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	13470700	mm ²
Platform Weight	24616	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1736	mm
Platform Height	1200	mm
Platform Clearance or End Offset	230	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	8000	mm
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1700	mm
Liquid Density	0.0006299	kg/cm ³
Element From Node	20	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	90	mm
Density	0.40000E-04	kg/cm ³
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N18"	
Dist. from "FROM" Node / Offset dist	260	mm
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	741.75	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N42"	
Dist. from "FROM" Node / Offset dist	150	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	119.23	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	20	



Detail Type Nozzle
Detail ID K3A3"
Dist. from "FROM" Node / Offset dist 1950 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 54.827702
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 119.23 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 20
Detail Type Nozzle
Detail ID K3B3"
Dist. from "FROM" Node / Offset dist 1950 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 125.172
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 119.23 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 20
Detail Type Nozzle
Detail ID K3C3"
Dist. from "FROM" Node / Offset dist 2250 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 54.827702
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 119.23 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 20
Detail Type Nozzle
Detail ID K3D3"
Dist. from "FROM" Node / Offset dist 2250 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 125.172
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 119.23 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 20
Detail Type Nozzle
Detail ID K3E3"
Dist. from "FROM" Node / Offset dist 2550 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 54.827702
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 119.23 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 20
Detail Type Nozzle
Detail ID K3F3"
Dist. from "FROM" Node / Offset dist 2550 mm
Nozzle Diameter 3 in.
Nozzle Schedule None



Nozzle Class 300
 Layout Angle 125.172
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 119.23 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID K61"
 Dist. from "FROM" Node / Offset dist 530 mm
 Nozzle Diameter 1 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID N52"
 Dist. from "FROM" Node / Offset dist 450 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 119.23 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 20
 Detail Type Weight
 Detail ID Trunnion-1
 Dist. from "FROM" Node / Offset dist 300 mm
 Miscellaneous Weight 735.45 N
 Offset from Element Centerline 850 mm

Element From Node 20
 Detail Type Weight
 Detail ID Trunnion-2
 Dist. from "FROM" Node / Offset dist 300 mm
 Miscellaneous Weight 735.45 N
 Offset from Element Centerline 850 mm

Element From Node 20
 Detail Type Weight
 Detail ID Deflector Plate
 Dist. from "FROM" Node / Offset dist 500 mm
 Miscellaneous Weight 196.12 N
 Offset from Element Centerline 200 mm

Element# 3/6
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description Shell-2
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 1700 mm
 Element Thickness 18 mm
 Internal Corrosion Allowance 1.5 mm
 Nominal Thickness 18 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 2.6002 MPa
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 0.1034 MPa



Design Temperature External Pressure 80 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 70
Efficiency, Longitudinal Seam 1.0
Efficiency, Circumferential Seam 1.0
Weld is pre-Heated No

Element From Node 30
Detail Type Liquid
Detail ID Liquid: 30
Dist. from "FROM" Node / Offset dist 0 mm
Height/Length of Liquid 1700 mm
Liquid Density 0.0006299 kg/cm³

Element From Node 30
Detail Type Insulation
Detail ID Ins: 30
Dist. from "FROM" Node / Offset dist 0 mm
Height/Length of Insulation 3000 mm
Thickness of Insulation 90 mm
Density 0.40000E-04 kg/cm³

Element From Node 30
Detail Type Nozzle
Detail ID K1A3"
Dist. from "FROM" Node / Offset dist 1650 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 54.827702
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 119.23 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K1B3"
Dist. from "FROM" Node / Offset dist 1650 mm
Nozzle Diameter 3 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 125.172
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 119.23 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K2A1.5"
Dist. from "FROM" Node / Offset dist 1450 mm
Nozzle Diameter 1.5 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 54.827702
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 30
Detail Type Nozzle
Detail ID K2B1.5"
Dist. from "FROM" Node / Offset dist 1450 mm
Nozzle Diameter 1.5 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 125.172
Blind Flange (Y/N) N



Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	30	
Detail Type	Nozzle	
Detail ID	K51.5"	
Dist. from "FROM" Node / Offset dist	2050	mm
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	103.322	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	30	
Detail Type	Nozzle	
Detail ID	K41.5"	
Dist. from "FROM" Node / Offset dist	1850	mm
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	103.322	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	81.556	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	30	
Detail Type	Weight	
Detail ID	Cleat weight	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	1569	N
Offset from Element Centerline	0	mm

Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	1850 mm
Inside Diameter	1700 mm
Element Thickness	18 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.6002 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Right Saddle
Dist. from "FROM" Node / Offset dist	400 mm
Width of Saddle	125 mm
Height of Saddle at Bottom	1180 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	180 mm
Thickness of Wear Plate	6 mm



Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1700	mm
Liquid Density	0.0006299	kg/cm ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1850	mm
Thickness of Insulation	90	mm
Density	0.40000E-04	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N36"	
Dist. from "FROM" Node / Offset dist	1640	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	454.21	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K78"	
Dist. from "FROM" Node / Offset dist	950	mm
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	638.2	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	Boot	
Dist. from "FROM" Node / Offset dist	1580	mm
Nozzle Diameter	12	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	920.68	N
Grade of Attached Flange	None	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	40	
Detail Type	Weight	
Detail ID	Demister	
Dist. from "FROM" Node / Offset dist	1580	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	1050	mm
Element From Node	40	



Detail Type	Weight	
Detail ID	PSV WEIGHT	
Dist. from "FROM" Node / Offset dist	950	mm
Miscellaneous Weight	29418	N
Offset from Element Centerline	2000	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Trunnion-3	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	735.45	N
Offset from Element Centerline	850	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Trunnion-4	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	735.45	N
Offset from Element Centerline	850	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Vortex breaker	
Dist. from "FROM" Node / Offset dist	1640	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	550	mm

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Elliptical
Description	Right Head
Distance "FROM" to "TO"	50 mm
Inside Diameter	1700 mm
Element Thickness	18 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.6002 MPa
Design Temperature Internal Pressure	80 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1700 mm
Liquid Density	0.0006299 kg/cm³

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	475 mm
Thickness of Insulation	90 mm
Density	0.40000E-04 kg/cm³

Element From Node	50
Detail Type	Nozzle
Detail ID	MH224"
Dist. from "FROM" Node / Offset dist	0 mm



Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	7833	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

Element From Node	50	
Detail Type	Weight	
Detail ID	Manway Davit	
Dist. from "FROM" Node / Offset dist	425	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	0	mm

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	Sump Head	
Distance "FROM" to "TO"	50	mm
Inside Diameter	300	mm
Element Thickness	12	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.6002	MPa
Design Temperature Internal Pressure	80	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 60	
Dist. from "FROM" Node / Offset dist	-75	mm
Height/Length of Insulation	125	mm
Thickness of Insulation	90	mm
Density	0.40000E-04	kg/cm³

Element From Node	60	
Detail Type	Nozzle	
Detail ID	N22"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	117.6	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2	+IT

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Left Head		50	...	50	...
Shell-1		3050	...	3000	...
Shell-2		6050	...	3000	...
Shell-3		7900	...	1850	...
Right Head		7950	...	50	...
Sump Head		7630	...	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Left Head		2.6107	22	1.5	1700	138	1
Shell-1		2.6107	18	1.5	1700	138	2
Shell-2		2.6107	18	1.5	1700	138	3
Shell-3		2.6107	18	1.5	1700	138	4
Right Head		2.6107	22	1.5	1700	138	5
Sump Head		2.6002	16	1.5	300	138	6

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Left Head		2.60015	2.66	2.92	18	17.6016	1
Shell-1		2.60015	2.63	2.89	18	17.7939	2
Shell-2		2.60015	2.63	2.89	18	17.7939	3
Shell-3		2.60015	2.63	2.89	18	17.7939	4
Right Head		2.60015	2.66	2.92	18	17.6016	5
Sump Head		2.60015	9.62	11	12	4.32275	6

Minimum

2.62

2.89

MAWP: 2.62 MPa, limited by: Nozzle Reinforcement.

Elements Suitable for Design Internal Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Left Head

Elliptical Head Design Information:					
Design Pressure	P	2.611 MPa	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1700.000 mm	Actual Stress		134.7 MPa
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		3570928.0 mm²	SG of Contents		0.631
Element Volume		756.7 ltr	Weight of Contents		4678.8 N
Empty Weight		6386.3 N	Operating Weight		11065.1 N
			Comp. All. Ope. Stress		129.0 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.625 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.61 \cdot 1703 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.61)$$

$$= 16.1016 + 1.5000 = 17.6016 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.011 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 16.5) / (1 \cdot 1703 + 0.2 \cdot 16.5)$$

$$= 2.68 - 0.011 = 2.66 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 18) / (1 \cdot 1700 + 0.2 \cdot 18)$$

$$= 2.92 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.61 \cdot (1 \cdot 1703 + 0.2 \cdot 16.5)) / (2 \cdot 1 \cdot 16.5)$$

$$= 134.672 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2.61 \cdot 852) / (138 \cdot 1 - 0.6 \cdot 2.61) + 1.5$$

$$= 17.794 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.011 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20.5) / (852 + 0.6 \cdot 20.5)$$

$$= 3.28 - 0.011 = 3.26 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$



$$= (2 + (1703 / (2 \cdot 427))^2) / 6$$

$$= 0.997657$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.500 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 18, tr = 16.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -11 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 22, tr = 16.4, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.8$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -16 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.628 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	22.000 mm	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.800	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-16 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.625 mm

EXTERNAL PRESSURE RESULTS for: Left Head

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
16.500	1736.00	105.21	0.0013201	93.38

$$MAEP = B / (K0 \cdot Do / t) = 93.4 / (0.9 \cdot 105) = 0.99 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
4.495	1736.00	386.22	0.0003596	35.95

$$MAEP = B / (K0 \cdot Do / t) = 36 / (0.9 \cdot 386) = 0.1 \text{ MPa}$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	2.611 MPa	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1700.000 mm	Actual Stress		136.3 MPa
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		16361414.0 mm ²	SG of Contents		0.631
Element Volume		6810.6 ltr	Weight of Contents		42108.9 N
Empty Weight		22150.7 N	Operating Weight		64259.6 N
			Comp. All. Ope. Stress		129.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.61 \cdot 852) / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 16.2939 + 1.5000 = 17.7939 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.011 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16.5) / (852 + 0.6 \cdot 16.5)$$

$$= 2.64 - 0.011 = 2.63 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 18) / (850 + 0.6 \cdot 18)$$

$$= 2.89 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.61 \cdot (852 + 0.6 \cdot 16.5)) / (1 \cdot 16.5)$$

$$= 136.293 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.048 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 18, t_r = 16.4, c = 1.5 mm, E* = 1

Thickness Ratio = t_r * E* / (t_g - c) = 0.99, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -11 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.628 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C



Stress/Thk. Ratio	0.994	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-11 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
16.500	1736.00	8233.33	105.21	4.7427	0.0002364	23.63

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.6)/(3*105) = 0.3 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.692	1736.00	8233.33	162.36	4.7427	0.0001260	12.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*12.6)/(3*162) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
16.500	1736.00	167790.48	105.21	50.0000	0.0001006	10.05

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.1)/(3*105) = 0.13 \text{ MPa}$$

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	2.611 MPa	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	80 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1700.000 mm	Actual Stress	136.3 MPa	
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.	18.000 mm	
Surface Area		16361414.0 mm ²	SG of Contents	0.631	
Element Volume		6810.6 ltr	Weight of Contents	42108.9 N	
Empty Weight		22150.7 N	Operating Weight	64259.6 N	
			Comp. All. Ope. Stress	129.0 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.61 \cdot 852) / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 16.2939 + 1.5000 = 17.7939 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.011 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16.5) / (852 + 0.6 \cdot 16.5)$$

$$= 2.64 - 0.011 = 2.63 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 18) / (850 + 0.6 \cdot 18)$$

$$= 2.89 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.61 \cdot (852 + 0.6 \cdot 16.5)) / (1 \cdot 16.5)$$

$$= 136.293 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.048 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 18, t_r = 16.4, c = 1.5 mm, E* = 1

Thickness Ratio = t_r * E* / (t_g - c) = 0.99, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -11 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.628 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C



Stress/Thk. Ratio	0.994	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-11 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
16.500	1736.00	8233.33	105.21	4.7427	0.0002364	23.63

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.6)/(3*105) = 0.3 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.692	1736.00	8233.33	162.36	4.7427	0.0001260	12.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*12.6)/(3*162) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
16.500	1736.00	167790.48	105.21	50.0000	0.0001006	10.05

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.1)/(3*105) = 0.13 \text{ MPa}$$

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	2.611 MPa	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	80 °C	
Length		1850.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1700.000 mm	Actual Stress	136.3 MPa	
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.	18.000 mm	
Surface Area		10089540.0 mm ²	SG of Contents	0.631	
Element Volume		4199.9 ltr	Weight of Contents	25967.1 N	
Empty Weight		13659.6 N	Operating Weight	39626.7 N	
			Comp. All. Ope. Stress	129.0 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.61 \cdot 852) / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 16.2939 + 1.5000 = 17.7939 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.011 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16.5) / (852 + 0.6 \cdot 16.5)$$

$$= 2.64 - 0.011 = 2.63 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 18) / (850 + 0.6 \cdot 18)$$

$$= 2.89 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.61 \cdot (852 + 0.6 \cdot 16.5)) / (1 \cdot 16.5)$$

$$= 136.293 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.048 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 18, t_r = 16.4, c = 1.5 mm, E* = 1

Thickness Ratio = t_r * E* / (t_g - c) = 0.99, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -11 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.628 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C



Stress/Thk. Ratio	0.994	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-11 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
16.500	1736.00	8233.33	105.21	4.7427	0.0002364	23.63

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.6)/(3*105) = 0.3 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.692	1736.00	8233.33	162.36	4.7427	0.0001260	12.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*12.6)/(3*162) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
16.500	1736.00	167790.48	105.21	50.0000	0.0001006	10.05

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.1)/(3*105) = 0.13 \text{ MPa}$$

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Right Head

Elliptical Head Design Information:					
Design Pressure	P	2.611 MPa	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	1700.000 mm	Actual Stress		134.7 MPa
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		3570928.0 mm²	SG of Contents		0.631
Element Volume		756.7 ltr	Weight of Contents		4678.8 N
Empty Weight		6386.3 N	Operating Weight		11065.2 N
			Comp. All. Ope. Stress		129.0 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.250 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.625 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.61 \cdot 1703 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.61)$$

$$= 16.1016 + 1.5000 = 17.6016 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.011 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 16.5) / (1 \cdot 1703 + 0.2 \cdot 16.5)$$

$$= 2.68 - 0.011 = 2.66 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 18) / (1 \cdot 1700 + 0.2 \cdot 18)$$

$$= 2.92 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.61 \cdot (1 \cdot 1703 + 0.2 \cdot 16.5)) / (2 \cdot 1 \cdot 16.5)$$

$$= 134.672 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2.61 \cdot 852) / (138 \cdot 1 - 0.6 \cdot 2.61) + 1.5$$

$$= 17.794 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.011 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20.5) / (852 + 0.6 \cdot 20.5)$$

$$= 3.28 - 0.011 = 3.26 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$



$$= (2 + (1703 / (2 \cdot 427))^2) / 6$$

$$= 0.997657$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.500 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 18, tr = 16.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -11 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 22, tr = 16.4, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.8$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -16 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.628 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	22.000 mm	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.800	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-16 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 21.250 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 10.625 mm

EXTERNAL PRESSURE RESULTS for: Right Head

Elliptical Head From 50 to 60 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
16.500	1736.00	105.21	0.0013201	93.38

$$MAEP = B / (K0 \cdot Do / t) = 93.4 / (0.9 \cdot 105) = 0.99 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
4.495	1736.00	386.22	0.0003596	35.95

$$MAEP = B / (K0 \cdot Do / t) = 36 / (0.9 \cdot 386) = 0.1 \text{ MPa}$$



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Sump Head

Elliptical Head Design Information:					
Design Pressure	P	2.600 MPa	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	300.000 mm	Actual Stress		37.3 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		171631.6 mm ²	SG of Contents		0.000
Element Volume		7.1 ltr	Weight of Contents		0.0 N
Empty Weight		205.7 N	Operating Weight		205.7 N
			Comp. All. Ope. Stress		129.0 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 3.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.875 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.6 \cdot 303 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.6)$$

$$= 2.8228 + 1.5000 = 4.3228 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 10.5) / (0.99 \cdot 303 + 0.2 \cdot 10.5)$$

$$= 9.62 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 12) / (1 \cdot 300 + 0.2 \cdot 12)$$

$$= 11 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.6 \cdot (0.99 \cdot 303 + 0.2 \cdot 10.5)) / (2 \cdot 1 \cdot 10.5)$$

$$= 37.288 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2.6 \cdot 152) / (138 \cdot 1 - 0.6 \cdot 2.6) + 1.5$$

$$= 4.387 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14.5) / (152 + 0.6 \cdot 14.5)$$

$$= 12.5 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (303 / (2 \cdot 76.5))^2) / 6$$

$$= 0.986992$$



Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 20.339 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 12$, $t_r = 2.84$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^*/(t_g - c) = 0.27$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 16$, $t_r = 2.91$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^*/(t_g - c) = 0.2$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2.617 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.271	Temperature Difference	81 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 3.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 1.875 mm

EXTERNAL PRESSURE RESULTS for: Sump Head

Elliptical Head From 60 to 70 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
10.500	324.00	30.86	0.0045010	118.10

$$MAEP = B/(K0 \cdot Do/t) = 118/(0.9 \cdot 30.9) = 4.25 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
0.839	324.00	386.22	0.0003596	35.95

$$MAEP = B/(K0 \cdot Do/t) = 36/(0.9 \cdot 386) = 0.1 \text{ MPa}$$



Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3.403 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3.380 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3.735 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	2.879 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.718 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.718 MPa
App 27-4	= M.A.W.P.	2.617 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 2.62 • 1
 = 3.403 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Left Head	161.8	234.0	0.691	3.42
Shell-1	163.5	234.0	0.699	3.42
Shell-2	163.5	234.0	0.699	3.42
Shell-3	163.5	234.0	0.699	3.42
Right Head	161.8	234.0	0.691	3.42
Sump Head	43.2	234.0	0.184	3.42

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
MH-1_24"	Nozzle	138.00	138.00	1.000
N1_8"	Nozzle	138.00	138.00	1.000
N4_2"	Nozzle	138.00	138.00	1.000
K3A_3"	Nozzle	138.00	138.00	1.000
K3B_3"	Nozzle	138.00	138.00	1.000
K3C_3"	Nozzle	138.00	138.00	1.000
K3D_3"	Nozzle	138.00	138.00	1.000
K3E_3"	Nozzle	138.00	138.00	1.000
K3F_3"	Nozzle	138.00	138.00	1.000
K6_1"	Nozzle	138.00	138.00	1.000
N5_2"	Nozzle	138.00	138.00	1.000
K1A_3"	Nozzle	138.00	138.00	1.000
K1B_3"	Nozzle	138.00	138.00	1.000
K2A_1.5"	Nozzle	138.00	138.00	1.000
K2B_1.5"	Nozzle	138.00	138.00	1.000
K5_1.5"	Nozzle	138.00	138.00	1.000
K4_1.5"	Nozzle	138.00	138.00	1.000
N3_6"	Nozzle	138.00	138.00	1.000
K7_8"	Nozzle	138.00	138.00	1.000
Boot	Nozzle	138.00	138.00	1.000
MH2_24"	Nozzle	138.00	138.00	1.000
N2_2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Left Head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000



Shell-3	138.00	138.00	1.000
Right Head	138.00	138.00	1.000
Sump Head	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
MH-1_24"	88.90	179.40	0.496
N1_8"	28.12	179.40	0.157
N4_2"	7.23	179.40	0.040
K3A_3"	8.38	179.40	0.047
K3B_3"	8.38	179.40	0.047
K3C_3"	8.38	179.40	0.047
K3D_3"	8.38	179.40	0.047
K3E_3"	8.38	179.40	0.047
K3F_3"	8.38	179.40	0.047
K6_1"	5.10	179.40	0.028
N5_2"	7.23	179.40	0.040
K1A_3"	8.38	179.40	0.047
K1B_3"	8.38	179.40	0.047
K2A_1.5"	6.12	179.40	0.034
K2B_1.5"	6.12	179.40	0.034
K5_1.5"	6.12	179.40	0.034
K4_1.5"	6.12	179.40	0.034
N3_6"	24.85	179.40	0.139
K7_8"	28.12	179.40	0.157
Boot	45.48	179.40	0.253
MH2_24"	88.90	179.40	0.496
N2_2"	7.24	179.40	0.040

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3.403 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3.380 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3.735 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	2.879 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.718 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.718 MPa
App 27-4	= M.A.W.P.	2.617 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 2.62 • 1
 = 3.403 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Left Head	176.4	234.0	0.754	3.42
Shell-1	178.5	234.0	0.763	3.42
Shell-2	178.5	234.0	0.763	3.42
Shell-3	178.5	234.0	0.763	3.42
Right Head	176.4	234.0	0.754	3.42
Sump Head	49.1	234.0	0.210	3.42

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
MH-1_24"	Nozzle	138.00	138.00	1.000
N1_8"	Nozzle	138.00	138.00	1.000
N4_2"	Nozzle	138.00	138.00	1.000
K3A_3"	Nozzle	138.00	138.00	1.000
K3B_3"	Nozzle	138.00	138.00	1.000
K3C_3"	Nozzle	138.00	138.00	1.000
K3D_3"	Nozzle	138.00	138.00	1.000
K3E_3"	Nozzle	138.00	138.00	1.000
K3F_3"	Nozzle	138.00	138.00	1.000
K6_1"	Nozzle	138.00	138.00	1.000
N5_2"	Nozzle	138.00	138.00	1.000
K1A_3"	Nozzle	138.00	138.00	1.000
K1B_3"	Nozzle	138.00	138.00	1.000
K2A_1.5"	Nozzle	138.00	138.00	1.000
K2B_1.5"	Nozzle	138.00	138.00	1.000
K5_1.5"	Nozzle	138.00	138.00	1.000
K4_1.5"	Nozzle	138.00	138.00	1.000
N3_6"	Nozzle	138.00	138.00	1.000
K7_8"	Nozzle	138.00	138.00	1.000
Boot	Nozzle	138.00	138.00	1.000
MH2_24"	Nozzle	138.00	138.00	1.000
N2_2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Left Head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000



Shell-3	138.00	138.00	1.000
Right Head	138.00	138.00	1.000
Sump Head	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
MH-1_24"	88.90	179.40	0.496
N1_8"	28.12	179.40	0.157
N4_2"	7.23	179.40	0.040
K3A_3"	8.38	179.40	0.047
K3B_3"	8.38	179.40	0.047
K3C_3"	8.38	179.40	0.047
K3D_3"	8.38	179.40	0.047
K3E_3"	8.38	179.40	0.047
K3F_3"	8.38	179.40	0.047
K6_1"	5.10	179.40	0.028
N5_2"	7.23	179.40	0.040
K1A_3"	8.38	179.40	0.047
K1B_3"	8.38	179.40	0.047
K2A_1.5"	6.12	179.40	0.034
K2B_1.5"	6.12	179.40	0.034
K5_1.5"	6.12	179.40	0.034
K4_1.5"	6.12	179.40	0.034
N3_6"	24.85	179.40	0.139
K7_8"	28.12	179.40	0.157
Boot	45.48	179.40	0.253
MH2_24"	88.90	179.40	0.496
N2_2"	7.24	179.40	0.040

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	1736	16.5	0.0013201	93.3796
20	30	8233.33	1736	16.5	0.0002364	23.6347
30	40	8233.33	1736	16.5	0.0002364	23.6347
40	50	8233.33	1736	16.5	0.0002364	23.6347
50	60	No Calc	1736	16.5	0.0013201	93.3796
60	70	No Calc	324	10.5	0.004501	118.105

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	18	5.99487	0.10343	0.99
20	30	18	12.1922	0.10343	0.3
30	40	18	12.1922	0.10343	0.3
40	50	18	12.1922	0.10343	0.3
50	60	18	5.99487	0.10343	0.99
60	70	12	2.3389	0.10343	4.25

Minimum

0.3

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	8233.33	167790	No Calc	No Calc
30	40	8233.33	167790	No Calc	No Calc
40	50	8233.33	167790	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	651.267	756.736	606.862	760.548	45.5887
20	30	2258.89	6810.62	2072.46	6834.68	158.122
30	40	2258.89	6810.62	2072.46	6834.68	158.122
40	50	1392.98	4199.88	1278.02	4214.72	97.5088
50	60	651.267	756.736	606.862	760.548	45.5887
60	70	20.976	7.06985	19.0095	7.24801	1.46832
Total		7234	19341.67	6655	19412.43	506

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	477.133	-141.667	0.72670E-04	...	Liquid: 10
10	Insl	17.8832	-187.5	...	...	Ins: 10
10	Nozl	854.712	-425	...	...	MH-1_24"
10	Wght	50	-425	...	...	Manway Davit
20	Sadl	147.252	1550	1003	...	Left Saddle
20	Plat	2685.97	4000	944.2	...	PLATFORM
20	Liqd	4294.2	1500	...	...	Liquid: 20
20	Insl	62.0273	1500	...	...	Ins: 20
20	Nozl	80.9373	260	946.838	...	N1_8"
20	Nozl	13.0095	150	-875.4	...	N4_2"
20	Nozl	13.0095	1950	511.579	725.954	K3A_3"
20	Nozl	13.0095	1950	-511.575	725.957	K3B_3"
20	Nozl	13.0095	2250	511.579	725.954	K3C_3"
20	Nozl	13.0095	2250	-511.575	725.957	K3D_3"
20	Nozl	13.0095	2550	511.579	725.954	K3E_3"
20	Nozl	13.0095	2550	-511.575	725.957	K3F_3"
20	Nozl	8.71379	530	862.7	...	K6_1"
20	Nozl	13.0095	450	-875.4	...	N5_2"
20	Wght	75	300	...	850	Trunnion-1
20	Wght	75	300	...	-850	Trunnion-2
20	Wght	20	500	200	...	Deflector Plate
30	Liqd	4294.2	1500	...	...	Liquid: 30
30	Insl	62.0273	1500	...	...	Ins: 30
30	Nozl	13.0095	1650	511.579	725.954	K1A_3"
30	Nozl	13.0095	1650	-511.575	725.957	K1B_3"
30	Nozl	8.89891	1450	500.605	710.382	K2A_1.5"
30	Nozl	8.89891	1450	-500.601	710.385	K2B_1.5"
30	Nozl	8.89891	2050	-200.249	845.664	K5_1.5"
30	Nozl	8.89918	1850	-200.249	845.664	K4_1.5"
30	Wght	160	1000	...	...	Cleat weight
40	Sadl	147.252	400	1003	...	Right Saddle
40	Liqd	2648.09	925	...	...	Liquid: 40
40	Insl	38.2502	925	...	...	Ins: 40
40	Nozl	49.5622	1640	-923.165	...	N3_6"
40	Nozl	69.6385	950	946.838	...	K7_8"
40	Nozl	122.906	1580	1002.4	...	Boot
40	Wght	10	1580	1050	...	Demister
40	Wght	3000	950	2000	...	PSV WEIGHT
40	Wght	75	1500	...	850	Trunnion-3
40	Wght	75	1500	...	-850	Trunnion-4
40	Wght	10	1640	-550	...	Vortex breaker
50	Liqd	477.141	191.667	0.72670E-04	...	Liquid: 50
50	Insl	17.8832	237.5	...	...	Ins: 50
50	Nozl	854.712	475	...	...	MH2_24"
50	Wght	50	475	...	...	Manway Davit
60	Insl	1.07862	...	-12.5	...	Ins: 60
60	Nozl	12.8324	...	-239.713	...	N2_2"



Total Weight of Each Detail Type:

Saddles	294.5
Platforms	2686.0
Liquid	12190.8
Insulation	199.1
Nozzles	2219.7
Weights	3600.0
Sum of the Detail Weights	21190.1 kgm

Note:

Because some nozzles were attached to a head or flange, their weights include the attached head or flange.

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	7718.2	7718.2	7718.2	7718.2	7718.2	7718.2
Saddles	294.5	294.5	294.5	294.5	294.5	294.5
Nozzles	2219.7	2219.7	2219.7	2219.7	2219.7	2219.7
Wld Weights	590.0	590.0	590.0	590.0	590.0	590.0
Platforms	...	...	2686.0	2686.0	2686.0	2686.0
Insulation	...	...	199.1	199.1	199.1	199.1
Empty Weights	...	...	...	3010.0	3010.0	...
Test Weights	...	3010.0	...	...	...	...
Ope Weights	...	...	...	...	...	3010.0
Ope. Liquid	...	...	...	...	...	12190.8
Test Liquid	...	19329.9	...	...	...	...
Totals	10822.4	33162.3	13707.6	16717.6	16717.6	28908.3

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	10822.4 kgm
Shop Test Wt.	- Fabricated weight + water (full)	33162.3 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	13707.6 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	16717.6 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	16717.6 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	28908.3 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	28291.3 kgm
Field Test Wt.	- Empty weight + test medium (full)	36047.4 kgm

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	3570928	3399753
20	30	16361414	16022123
30	40	16361414	16022123
40	50	10089540	9880310
50	60	3570928	3399753
60	70	171632	144683

Total 50125860.0 48868744.0 mm²

PV Elite 28 Licensee: SL User

FileName : V-31002_7%

Page 45 of 218

Element and Detail Weights:

Step: 13

1:47pm

May 20, 2026



HEXAGON

Total	50.1	48.9 m ²
-------	------	---------------------

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
MH-1_24"	4.79	5.10	80	300	GR 1.1	...	...	...	...
N1_8"	4.79	5.10	80	300	GR 1.1	2.764	4.427	4.794	4.427
N4_2"	9.59	10.20	80	600	GR 1.1	5.668	8.715	9.589	8.715
K3A_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K3B_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K3C_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K3D_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K3E_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K3F_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K6_1"	4.79	5.10	80	300	GR 1.1	...	...	...	...
N5_2"	9.59	10.20	80	600	GR 1.1	5.668	8.715	9.589	8.715
K1A_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K1B_3"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K2A_1.5"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K2B_1.5"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K5_1.5"	4.79	5.10	80	300	GR 1.1	...	...	...	...
K4_1.5"	4.79	5.10	80	300	GR 1.1	...	...	...	...
N3_6"	4.79	5.10	80	300	GR 1.1	3.171	4.020	4.794	4.020
K7_8"	4.79	5.10	80	300	GR 1.1	2.764	4.427	4.794	4.427
MH2_24"	4.79	5.10	80	300	GR 1.1	...	...	...	...
N2_2"	9.59	10.20	80	600	GR 1.1	5.668	8.715	9.589	8.715
Min Rating	4.020	5.100, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1_8"	248.539	0.50	8000000	5400	2.617	4.427	Pass
N4_2"	77.851	0.50	500000	1300	2.617	8.715	Pass
N5_2"	77.851	0.50	500000	1300	2.617	8.715	Pass
N3_6"	196.088	0.50	4500001	4000	2.617	4.020	Pass
K7_8"	248.539	0.50	8000000	5400	2.617	4.427	Pass
N2_2"	77.851	0.50	500000	1300	2.617	8.715	Pass

Evaluating Flange on Nozzle [N18"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(8000000) + 4 \cdot \text{abs}(5400)248.539 \leq \pi \cdot 248.539^3 \cdot ((4.794 - 2.617) + .500 \cdot 4.794)$$

$$= 0.13337E+09 \leq 0.22069E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N18" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.794, 0.500 \cdot 4.794 + 4.794 - 133370528.0 / (\pi \cdot 248.539^3))$$

$$= \min(4.794, 4.427)$$

$$= 4.427 \text{ MPa}$$

Evaluating Flange on Nozzle [N42"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(500000) + 4 \cdot \text{abs}(1300)77.851 \leq \pi \cdot 77.851^3 \cdot ((9.589 - 2.617) + .500 \cdot 9.589)$$

$$= 0.84050E+07 \leq 0.17446E+08 \text{ Check Passes}$$



Maximum Pressure for Flange on Nozzle N42" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(9.589, 0.500 \cdot 9.589 + 9.589 - 8404991.0 / (\pi \cdot 77.851^3)) \\ &= \min(9.589, 8.715) \\ &= 8.715 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N52"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(500000) + 4 \cdot \text{abs}(1300) 77.851 \leq \pi \cdot 77.851^3 [(9.589 - 2.617) + .500 \cdot 9.589] \\ &= 0.84050E+07 \leq 0.17446E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N52" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(9.589, 0.500 \cdot 9.589 + 9.589 - 8404991.0 / (\pi \cdot 77.851^3)) \\ &= \min(9.589, 8.715) \\ &= 8.715 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N36"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(4500001) + 4 \cdot \text{abs}(4000) 196.088 \leq \pi \cdot 196.088^3 [(4.794 - 2.617) + .500 \cdot 4.794] \\ &= 0.75139E+08 \leq 0.10838E+09 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N36" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4.794, 0.500 \cdot 4.794 + 4.794 - 75138680.0 / (\pi \cdot 196.088^3)) \\ &= \min(4.794, 4.020) \\ &= 4.020 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [K78"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(8000000) + 4 \cdot \text{abs}(5400) 248.539 \leq \pi \cdot 248.539^3 [(4.794 - 2.617) + .500 \cdot 4.794] \\ &= 0.13337E+09 \leq 0.22069E+09 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle K78" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4.794, 0.500 \cdot 4.794 + 4.794 - 133370528.0 / (\pi \cdot 248.539^3)) \\ &= \min(4.794, 4.427) \\ &= 4.427 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N22"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(500000) + 4 \cdot \text{abs}(1300) 77.851 \leq \pi \cdot 77.851^3 [(9.589 - 2.617) + .500 \cdot 9.589] \\ &= 0.84050E+07 \leq 0.17446E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N22" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(9.589, 0.500 \cdot 9.589 + 9.589 - 8404991.0 / (\pi \cdot 77.851^3)) \\ &= \min(9.589, 8.715) \\ &= 8.715 \text{ MPa} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 3.904

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50 • 1 • 1 • 1 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1984, 1 • 0.93 • 1 • 0.6 • 57.5²)

= 1843.5 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.92 • 0.76 • 1844

= 1290.7 N

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	1480.00	1.0000	1.0000	1.0000	57.50	0.7635
20	1480.00	1.0000	1.0000	1.0000	57.50	0.7635
30	1480.00	1.0000	1.0000	1.0000	57.50	0.7635
40	1480.00	1.0000	1.0000	1.0000	57.50	0.7635
50	1480.00	1.0000	1.0000	1.0000	57.50	0.7635
60	428.36	1.0000	1.0000	1.0000	57.50	0.7635

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLATFORM	13470700.00	1480.00	1.84	24842.37	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Manway Davit	0.00	1480.00	1.84	0.00
Trunnion-1	0.00	1480.00	1.84	0.00
Trunnion-2	0.00	1480.00	1.84	0.00
Deflector Pla	0.00	1680.00	1.84	0.00
Cleat weight	0.00	1480.00	1.84	0.00



Demister	0.00	2530.00	1.84	0.00
PSV WEIGHT	0.00	3480.00	1.84	0.00
Trunnion-3	0.00	1480.00	1.84	0.00
Trunnion-4	0.00	1480.00	1.84	0.00
Vortex breake	0.00	300.00	1.84	0.00
Manway Davit	0.00	1480.00	1.84	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1480	2299.2	917031	1.84422	1419.76
20	30	1480	2299.2	6897600	1.84422	38005.6
30	40	1480	2299.2	6897600	1.84422	10679
40	50	1480	2299.2	4253520	1.84422	6585.37
50	60	1480	2299.2	917031	1.84422	1419.76
60	70	428.364	604.8	114317	1.84422	176.987

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.094 \cdot 1) \cdot 1$
 $= 0.094$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.03 secs
Computed Sa/g = 0.094

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 39220 \cdot 0.094 \cdot 1$
 $= 3681.0 \text{ N}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.094 \cdot 274537 \cdot 1$
 $= 17177.9 \text{ N}$

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------



From	To	Height mm	Weight N	Ope Load N
10	20	850	39219.5	3680.99
20	30	850	39219.5	3680.99
20	30	850	39219.5	3680.99
30	40	850	39219.5	3680.99
40	0	850	39219.5	3680.99
40	50	850	39219.5	3680.99
50	60	850	39219.5	3680.99

The Earthquake Weight does not include the weight of the saddles.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

**Shop/Field Installation Options :**

Platform(s) installed in the Shop.

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4025.000 mm
Center of Gravity of the Platforms	4050.000 mm
Center of Gravity of the Liquid	3975.002 mm
Center of Gravity of the Insulation	3994.796 mm
Center of Gravity of the Nozzles	4148.929 mm
Center of Gravity of the Added Weights (Operating)	6498.527 mm
Center of Gravity of the Added Weights (Empty)	6498.527 mm
Center of Gravity of Bare Shell New and Cold	3985.570 mm
Center of Gravity of Bare Shell Corroded	3985.413 mm
Vessel CG in the Operating Condition	4319.914 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4558.793 mm
Vessel CG in the Test Condition	4246.652 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(125 + 1.56 \cdot \sqrt{ 860 \cdot 16.5 }, 2 \cdot 1550)$$

$$= 310.8030 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	859.75	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	7950.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1550.00	mm
Saddle Width	b1 or b	125.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	434.75	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1180.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	197201.14	N
Pressure used in Analysis	P	2.605 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	66.16	138.00
Long. Stress at Bottom of Midspan	69.60	138.00
Long. Stress at Top of Saddles	85.01	138.00
Long. Stress at Bottom of Saddles	58.39	138.00
Tangential Shear in Shell	9.25	110.40
Circ. Stress at Horn of Saddle	67.04	172.50
Circ. Compressive Stress in Shell	2.92	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (58286 / 2 + 0) \cdot 1180 / 1514$$

$$= 68150.5 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(13494, 0) \cdot 1180 / 4850$$

$$= 3283.0 \text{ N}$$

Saddle Reaction Force due to Earthquake FI [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(25767, 0) \cdot 1180 / 4850$$

$$= 6269.1 \text{ N}$$



Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (25767/2 + 0) \cdot 1180/1514$$

$$= 30127.5 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 129051 + \max(3283, 68150, 6269, 30128)$$

$$= 197201.1 \text{ N}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1844 (2.5 + \pi/4(1.92 \cdot 1.2)^2)$$

$$= 13494.322 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	198645.09	N
Transverse Shear Load	29143.23	N
Longitudinal Shear Load	25766.92	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1700/2 + 18 + 6) \sin(120/2), 2 \cdot 860)$$

$$= 1513.812 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -197201 \cdot 1550 [1 - (1 - 1550/7950 + (860^2 - 435^2)/(2 \cdot 1550 \cdot 7950)) / (1 + (4 \cdot 435)/(3 \cdot 7950))]$$

$$= -69985528.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 197201 \cdot 7950/4 ((1 + 2(860^2 - 435^2)/(7950^2)) / (1 + (4 \cdot 435)/(3 \cdot 7950)) - 4 \cdot 1550/7950)$$

$$= 66026364.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 2.61 \cdot 860/(2 \cdot 16.5) - 66026364/(\pi \cdot 860^2 \cdot 16.5)$$

$$= 66.16 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 2.61 \cdot 860/(2 \cdot 16.5) + 66026364/(\pi \cdot 860^2 \cdot 16.5)$$

$$= 69.60 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 2.61 \cdot 860/(2 \cdot 16.5) - 69985528/(0.11 \cdot \pi \cdot 860^2 \cdot 16.5)$$

$$= 85.01 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 2.61 \cdot 860/(2 \cdot 16.5) + 69985528/(0.19 \cdot \pi \cdot 860^2 \cdot 16.5)$$

$$= 58.39 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 197201 (7950 - 2 \cdot 1550) / (7950 + (4 \cdot 435/3))$$



= 112129.3 N

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 112129 / (860 \cdot 16.5)$$

$$= 9.25 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(860 \cdot 16.5)}$$

$$= 92.901 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 197201 \cdot 0.1 / (16.5(125 + 92.9 + 92.9))$$

$$= -2.92 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -197201 / (4 \cdot 16.5(125 + 92.9 + 92.9)) -$$

$$3 \cdot 0.053 \cdot 197201 / (2 \cdot 16.5^2)$$

$$= -67.04 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(125 + 1.56 \cdot \sqrt{(860 \cdot 16.5)}, 2 \cdot 1550)$$

$$= 310.80 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 4850.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.11973\text{E-}04 \cdot 4850(80 - 21.1)$$

$$= 3.419 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1575.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	243.0	MPa
Height of Web at Center	Hw,c	290.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(852 \cdot 16.5)} + 180)^3) / 12$$

$$= 136601.438 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 136601 + 6021(107 - 8.25)^2$$

$$= 58733384.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	364.9	16.5	8.3	6021.0	49673.3	0.587E+08
Wearplate	180.0	6.0	19.5	1080.0	21060.0	0.825E+07
Web	10.0	290.0	167.5	2900.0	485749.8	0.310E+08
BasePlate	150.0	16.0	320.5	2400.0	769199.8	0.110E+09
Totals	...	...	...	12401.0	1325682.8	0.208E+09



Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 1325682/12401 \\
 &= 106.901 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 197201 \\
 &= 40134.7070 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 6.2912 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 145.7982 \text{ MPa}
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1180 - 852 \cdot \sin(120/2) / 1.05 \\
 &= 475.815 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 19104440.0000 \text{ N-mm}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 9.8387 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 161.9980 \text{ MPa}
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt[3]{(3(197201 + 1444)150) / (2 \cdot 1575 \cdot 162)} \\
 &= 13.236 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (850 + 18 + 6) \\
 &= 1513.812 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 1514 / (3 - 1) \\
 &= 756.906 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 757 \cdot 150 / 2 \\
 &= 56767.965 \text{ mm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 198645 / (1575 \cdot 150) \\
 &= 0.841 \text{ N/mm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 56768 \cdot 0.84 \\
 &= 47731.969 \text{ N}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 1150 + 3785 \\
 &= 4934.531 \text{ mm}^2
 \end{aligned}$$

**Compressive Stress [Sc]:**

$$= P/A_r$$

$$= 47732/4935$$

$$= 9.674 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	115.0	67.5	1150.0	77625.0	0.391E+07
Web	378.5	10.0	5.0	3784.5	18922.7	0.834E+06
Totals	...	...	...	4934.5	96547.6	0.474E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 125 - 10$$

$$= 115.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 96548/4935$$

$$= 19.566 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(19.6, 125 - 19.6)$$

$$= 105.434 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4744205/4935)}$$

$$= 31.007 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1180 - \cos(120/2) (850 + 18 + 6) - 16$$

$$= 727.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/243)}$$

$$= 127.447$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 727/31$$

$$= 23.446$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 25767 / (2 \cdot 1575) \cdot 757 \cdot 727/2$$

$$= 2251513.500 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($23.4 < 127$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (23.4)^2 / (2 \cdot 127^2)) 243 / (5/3 + 3 \cdot (23.4) / (8 \cdot 127) - (23.4)^3 / (8 \cdot 127^3))$$

$$= 137.7 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 9.67/138 + (2251513 \cdot 105/4744210) / 162$$

$$= 0.379$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



Rib	10.0	115.0	67.5	1150.0	77625.0	0.465E+07
Web	756.9	10.0	5.0	7569.1	37845.3	0.577E+06
Totals	...	...	...	8719.1	115470.3	0.523E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 115470/8719$$

$$= 13.243 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(13.2, \text{abs}(125 - 13.2))$$

$$= 111.757 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1180 - \cos(874/2) (16 + 0 + 0) - 0$$

$$= 290.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (5230157/8719) }$$

$$= 24.492 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 290/24.5$$

$$= 11.841$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 25767 / (2 \cdot 1575)$$

$$= 8.180 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 8.18 \cdot 757 \cdot 290$$

$$= 1796255.375 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1796255/46800$$

$$= 38.370 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($11.8 < 127$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.8)^2 / (2 \cdot 127^2)) 243 / (5/3 + 3 \cdot (11.8) / (8 \cdot 127) - (11.8^3) / (8 \cdot 127^3))$$

$$= 142.2 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 11/142 + (1796255 \cdot 112/5230157) / 162$$

$$= 0.314$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.30 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	225.7742 mm²
Saddle Load Q0 (Weight)	Q0	130494.6 N



Saddle Load QL (Wind/Seismic contribution)	QL	6269.1 N
Maximum Transverse Force	Ft	29143.2 N
Maximum Longitudinal Force	Fl	25766.9 N
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
= 49292.2 N

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
= $\max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $49292 / (226 \cdot 4)$
= 54.6 MPa. Must be less than 107.8 MPa.

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $29143 / (226 \cdot 4)$
= 32.3 MPa. Must be less than 107.8 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $25767 / (108 \cdot 4)$
= 59.7614 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1180 \cdot 29143 + 0$
= 34402952.00 N-mm

Eccentricity (e):
= R_{mom} / QO
= $34402952 / 130495$
= 263.53 mm > Bplen/6 --> Uplift in Transverse direction

$f = B_{plen} / 2 - \text{Edgedis}$
= $1575 / 2 - 50.8$
= 736.70 mm

Modular Ratio Of Steel/Concrete (n1):
= E_S / E_C
= $203402 / 21526$
= 9.45

$K1 = 3 (e - 0.5 \cdot B_{plen})$
= $3 (264 - 0.5 \cdot 1575)$
= -1571.92 mm

$K2 = 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$
= $6 \cdot 9.45 \cdot 452 / 150 \cdot (737 + 264)$
= 170706.42 mm²

$K3 = -K2 \cdot (0.5 \cdot B_{plen} + f)$
= $-170706 \cdot (0.5 \cdot 1575 + 737)$
= -260190708.62 mm³

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -1572 \cdot Y^2 + 170706 \cdot Y + -260190704 = 0$
 $Y = 1568.82 \text{ mm}$

$\text{Num} = (B_{plen} / 2 - Y / 3 - e)$
= $(1575 / 2 - 1569 / 3 - 264)$
= 1.03



$$\begin{aligned}\text{Denom} &= (\text{Bplen} / 2 - Y / 3 + f) \\ &= (1575/2 - 1569/3 + 737) \\ &= 1001.26\end{aligned}$$

$$\begin{aligned}\text{Total Bolt Tension Force [Tforce]:} \\ &= -QO \cdot \text{Num} / \text{Denom} \\ &= -130495 \cdot 1.03/1001 \\ &= -134.46 \text{ N}\end{aligned}$$

$$\begin{aligned}\text{Bolt Area Required due to Transverse Load [Bltareart]:} \\ &= \text{Tforce} / (\text{Stba} \cdot \text{Nbt}) \\ &= -134 / (137 \cdot 2) \\ &= -0.4897 \text{ mm}^2\end{aligned}$$

$$\begin{aligned}\text{Required Area of a Single Bolt [Bltarear]:} \\ &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\ &= \max[0, 59.8, -0.49] \\ &= 59.7614 \text{ mm}^2\end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned}\text{Bearing Pressure (fc)} \\ &= 2(QO + \text{Tforce}) / (Y \cdot \text{Bpwid}) \\ &= 2(130495 + -134) / (1569 \cdot 150) \\ &= 1.11 \text{ MPa}\end{aligned}$$

$$\begin{aligned}\text{Distance from Baseplate Edge to the Web [ADIST]:} \\ &= (\text{Bplen} - \text{WebLength}) / 2 \\ &= (1575 - 1524) / 2 \\ &= 25.4000 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Overturning Moment due To Bolt Tension [Mt]:} \\ &= \text{Tforce} \cdot \text{Adist} \\ &= -134 \cdot 25.4 \\ &= -3416.54 \text{ N-mm}\end{aligned}$$

$$\begin{aligned}\text{Equivalent Bearing Pressure (f1):} \\ &= \text{fc} \cdot (Y - \text{Adist}) / Y \\ &= 1.11 \cdot (1569 - 25.4) / 1569 \\ &= 1.09 \text{ MPa}\end{aligned}$$

$$\begin{aligned}\text{Overturning Moment due to Bearing Pressure [Mc]:} \\ &= (\text{Adist}^2 \cdot \text{Bpwid} / 6) \cdot (f1 + 2 \cdot \text{fc}) \\ &= (25.4^2 \cdot 150 / 6) \cdot (1.09 + 2 \cdot 1.11) \\ &= 53341.47 \text{ N-mm}\end{aligned}$$

$$\begin{aligned}\text{Baseplate Required Thickness [Treq]:} \\ &= \sqrt{(6 \cdot \max(\text{Mt}, \text{Mc}) / (\text{Bpwid} \cdot \text{Sba}))} \\ &= \sqrt{(6 \cdot \max(-3417, 53341) / (150 \cdot 207))} \\ &= 3.2101 \text{ mm}\end{aligned}$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 6$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

$$\begin{aligned}\text{Minimum Wear Plate Width to be considered in analysis [b1]:} \\ &= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)\end{aligned}$$



$$= \min(125 + 1.56\sqrt{(860 \cdot 16.5)}, 2 \cdot 1550)$$

$$= 310.8030 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	859.75	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	7950.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1550.00	mm
Saddle Width	b1 or b	125.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	434.75	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1180.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		231013.77	N
Pressure used in Analysis	P	2.605	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	65.86	138.00
Long. Stress at Bottom of Midspan	69.90	138.00
Long. Stress at Top of Saddles	87.94	138.00
Long. Stress at Bottom of Saddles	56.76	138.00
Tangential Shear in Shell	10.84	110.40
Circ. Stress at Horn of Saddle	78.54	172.50
Circ. Compressive Stress in Shell	3.43	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (58286/2 + 0) \cdot 1180/1514$$

$$= 68150.5 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(13494, 0) \cdot 1180/4850$$

$$= 3283.0 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(25767, 0) \cdot 1180/4850$$

$$= 6269.1 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (25767/2 + 0) \cdot 1180/1514$$

$$= 30127.5 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 162863 + \max(3283, 68150, 6269, 30128)$$

$$= 231013.8 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$



$$= 1.1 \cdot 1844 (2.5 + \pi/4 (1.92 \cdot 1.2)^2)$$

$$= 13494.322 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	232457.70	N
Transverse Shear Load	29143.23	N
Longitudinal Shear Load	25766.92	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1700/2 + 18 + 6)\sin(120/2), 2 \cdot 860)$$

$$= 1513.812 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -231014 \cdot 1550 [1 - (1 - 1550/7950 + (860^2 - 435^2)/(2 \cdot 1550 \cdot 7950))/(1 + (4 \cdot 435)/(3 \cdot 7950))]$$

$$= -81985432.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 231014 \cdot 7950/4 [(1 + 2(860^2 - 435^2)/(7950^2))/(1 + (4 \cdot 435)/(3 \cdot 7950)) - 4 \cdot 1550/7950]$$

$$= 77347424.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 2.61 \cdot 860/(2 \cdot 16.5) - 77347424/(\pi \cdot 860^2 \cdot 16.5)$$

$$= 65.86 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 2.61 \cdot 860/(2 \cdot 16.5) + 77347424/(\pi \cdot 860^2 \cdot 16.5)$$

$$= 69.90 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2.61 \cdot 860/(2 \cdot 16.5) - 81985432/(0.11 \cdot \pi \cdot 860^2 \cdot 16.5)$$

$$= 87.94 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2.61 \cdot 860/(2 \cdot 16.5) + 81985432/(0.19 \cdot \pi \cdot 860^2 \cdot 16.5)$$

$$= 56.76 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 231014 (7950 - 2 \cdot 1550)/(7950 + (4 \cdot 435/3))$$

$$= 131355.3 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 131355/(860 \cdot 16.5)$$

$$= 10.84 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(860 \cdot 16.5)}$$

$$= 92.901 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:



$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 231014 \cdot 0.1 / (16.5 (125 + 92.9 + 92.9))$$

$$= -3.43 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -231014 / (4 \cdot 16.5 (125 + 92.9 + 92.9)) -$$

$$3 \cdot 0.053 \cdot 231014 / (2 \cdot 16.5^2)$$

$$= -78.54 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(125 + 1.56 \cdot \sqrt{860 \cdot 16.5}, 2 \cdot 1550)$$

$$= 310.80 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1575.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	243.0	MPa
Height of Web at Center	Hw,c	290.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{852 \cdot 16.5}) + 180)^3 / 12$$

$$= 136601.438 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 136601 + 6021 (107 - 8.25)^2$$

$$= 58733384.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	364.9	16.5	8.3	6021.0	49673.3	0.587E+08
Wearplate	180.0	6.0	19.5	1080.0	21060.0	0.825E+07
Web	10.0	290.0	167.5	2900.0	485749.8	0.310E+08
BasePlate	150.0	16.0	320.5	2400.0	769199.8	0.110E+09
Totals	...	...	...	12401.0	1325682.8	0.208E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1325682 / 12401$$

$$= 106.901 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 231014$$

$$= 47016.3125 \text{ N}$$



Tension Stress, $St = (Fh/As) = 7.3700$ MPa
 Allowed Stress, $Sa = 0.6 * Yield Str = 145.7982$ MPa

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians})$
 $= 1180 - 852 \cdot \sin(120/2)/1.05$
 $= 475.815$ mm

Bending Moment, $M = Fh * d = 22380138.0000$ N-mm

Bending Stress, $Sb = (M * C1 / I) = 11.5257$ MPa
 Allowed Stress, $Sa = 2/3 * Yield Str = 161.9980$ MPa

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + SaddleWt)BasePlateWidth / (2 * BasePlateLength * AllStress)}$
 $= \sqrt{3(231014 + 1444)150 / (2 * 1575 * 162)}$
 $= 14.318$ mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)$
 $= 2 \cdot \cos(90 - 120/2)(850 + 18 + 6)$
 $= 1513.812$ mm

Distance between Ribs [e]:
 $= Web Length / (Nribs - 1)$
 $= 1514 / (3 - 1)$
 $= 756.906$ mm

Baseplate Pressure Area [Ap]:
 $= e \cdot Bpwid / 2$
 $= 757 \cdot 150 / 2$
 $= 56767.965$ mm²

Bearing Pressure [Bp]:
 $= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$
 $= 232458 / (1575 \cdot 150)$
 $= 0.984$ N/mm²

Axial Load [P]:
 $= Ap \cdot Bp$
 $= 56768 \cdot 0.98$
 $= 55856.727$ N

Area of the Rib and Web [Ar]:
 $= Rib Area + Web Area$
 $= 1150 + 3785$
 $= 4934.531$ mm²

Compressive Stress [Sc]:
 $= P/Ar$
 $= 55857/4935$
 $= 11.321$ MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	115.0	67.5	1150.0	77625.0	0.391E+07
Web	378.5	10.0	5.0	3784.5	18922.7	0.834E+06
Totals	...	...	...	4934.5	96547.6	0.474E+07

Rib dimension [D]:
 $= Saddle Width - Web Thickness$
 $= 125 - 10$
 $= 115.000$ mm



Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 96548/4935$$

$$= 19.566 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{Saddle Width} - ytot)$$

$$= \max(19.6, 125 - 19.6)$$

$$= 105.434 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 4744205/4935 }$$

$$= 31.007 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1180 - \cos(120/2) (850 + 18 + 6) - 16$$

$$= 727.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199955/243) }$$

$$= 127.447$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 727/31$$

$$= 23.446$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 25767 / (2 \cdot 1575) \cdot 757 \cdot 727/2$$

$$= 2251513.500 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($23.4 < 127$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.4)^2 / (2 \cdot 127^2)) 243 / (5/3 + 3 \cdot (23.4) / (8 \cdot 127) - (23.4^3) / (8 \cdot 127^3))$$

$$= 137.7 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.3/138 + (2251513 \cdot 105/4744210) / 162$$

$$= 0.391$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	115.0	67.5	1150.0	77625.0	0.465E+07
Web	756.9	10.0	5.0	7569.1	37845.3	0.577E+06
Totals	...	...	...	8719.1	115470.3	0.523E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 115470/8719$$

$$= 13.243 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(13.2, \text{abs}(125 - 13.2))$$

$$= 111.757 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1180 - \cos(874/2) (16 + 0 + 0) - 0$$

$$= 290.000 \text{ mm}$$



Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{5230157/8719}$$

$$= 24.492 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 290/24.5$$

$$= 11.841$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 25767 / (2 \cdot 1575)$$

$$= 8.180 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 8.18 \cdot 757 \cdot 290$$

$$= 1796255.375 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1796255/46800$$

$$= 38.370 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($11.8 < 127$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.8)^2 / (2 \cdot 127^2)) 243 / (5/3 + 3 \cdot (11.8) / (8 \cdot 127) - (11.8)^3 / (8 \cdot 127^3))$$

$$= 142.2 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 12.8/142 + (1796255 \cdot 112/5230157) / 162$$

$$= 0.327$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.30 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BlArea	225.7742 mm ²
Saddle Load Q0 (Weight)	Q0	164307.2 N
Saddle Load QL (Wind/Seismic contribution)	QL	6269.1 N
Maximum Transverse Force	Ft	29143.2 N
Maximum Longitudinal Force	F1	25766.9 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 25767 / (226 \cdot 4)$$

$$= 28.5 \text{ MPa. Must be less than } 107.8 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 29143 / (226 \cdot 4)$$

$$= 32.3 \text{ MPa. Must be less than } 107.8 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$



Bolt Area due to Shear Load [Bltarears]:

= $F_t / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$

= $25767 / (108 \cdot 4)$

= 59.7614 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= $1180 \cdot 29143 + 0$

= 34402952.00 N-mm

Eccentricity (e):

= R_{mom} / QO

= $34402952 / 164307$

= 209.30 mm < $B_{\text{plen}}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$

= $\max[0, 59.8, 0]$

= 59.7614 mm²

MDMT Calculations for: Right Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 6$, $c = 1.5$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -29 °C

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(125 + 1.56 \cdot \sqrt{ 859 \cdot 18 }, 2 \cdot 1550)$$

$$= 318.9802 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	859.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	7950.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1550.00	mm
Saddle Width	b1 or b	125.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	434.00	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1180.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		181699.50	N
Pressure used in Analysis	P	3.411	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	79.93	234.00
Long. Stress at Bottom of Midspan	82.85	234.00
Long. Stress at Top of Saddles	95.88	234.00
Long. Stress at Bottom of Saddles	73.36	234.00
Tangential Shear in Shell	7.82	138.58
Circ. Stress at Horn of Saddle	52.37	350.99
Circ. Compressive Stress in Shell	2.41	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (19235 / 2 + 0) \cdot 1180 / 1514$$

$$= 22489.7 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(4453, 0) \cdot 1180 / 4850$$

$$= 1083.4 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 159210 + \max(1083, 22490, 0, 0)$$

$$= 181699.5 \text{ N}$$



Longitudinal Wind Force [F_L]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1844 (2.5 + \pi/4 (1.92 \cdot 1.2)^2)$
 $= 13494.322 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	183143.44	N
Transverse Shear Load	9617.27	N
Longitudinal Shear Load	4452.90	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(1700/2 + 18 + 6) \sin(120/2), 2 \cdot 859)$
 $= 1513.812 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -181700 \cdot 1550 [1 - (1 - 1550/7950 + (859^2 - 434^2)/(2 \cdot 1550 \cdot 7950)) / (1 + (4 \cdot 434)/(3 \cdot 7950))]$
 $= -64465412.0 \text{ N-mm}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/(3L)) - 4a/L)$
 $= 181700 \cdot 7950/4 ((1 + 2(859^2 - 434^2)/(7950^2)) / (1 + (4 \cdot 434)/(3 \cdot 7950)) - 4 \cdot 1550/7950)$
 $= 60869480.0 \text{ N-mm}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 3.41 \cdot 859/(2 \cdot 18) - 60869480/(\pi \cdot 859^2 \cdot 18)$
 $= 79.93 \text{ MPa}$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 3.41 \cdot 859/(2 \cdot 18) + 60869480/(\pi \cdot 859^2 \cdot 18)$
 $= 82.85 \text{ MPa}$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$
 $= 3.41 \cdot 859/(2 \cdot 18) - 64465412/(0.11 \cdot \pi \cdot 859^2 \cdot 18)$
 $= 95.88 \text{ MPa}$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 $= P \cdot R_m/(2t) + M_1/(K_1^* \cdot \pi \cdot R_m^2 \cdot t)$
 $= 3.41 \cdot 859/(2 \cdot 18) + 64465412/(0.19 \cdot \pi \cdot 859^2 \cdot 18)$
 $= 73.36 \text{ MPa}$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L - 2a)/(L + (4 \cdot h_m/3))$
 $= 181700 (7950 - 2 \cdot 1550) / (7950 + (4 \cdot 434/3))$
 $= 103327.1 \text{ N}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K_2 \cdot T / (R_m \cdot t)$
 $= 1.17 \cdot 103327 / (859 \cdot 18)$
 $= 7.82 \text{ MPa}$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{ R_m \cdot t }$
 $= 0.78 \cdot \sqrt{ 859 \cdot 18 }$



= 96.990 mm

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 181700 \cdot 0.1 / (18(125 + 97 + 97))$$

$$= -2.41 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -181700 / (4 \cdot 18(125 + 97 + 97)) - 3 \cdot 0.053 \cdot 181700 / (2 \cdot 18^2)$$

$$= -52.37 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(125 + 1.56 \cdot \sqrt{859 \cdot 18}, 2 \cdot 1550)$$

$$= 318.98 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1575.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	290.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{850 \cdot 18} + 180)18^3) / 12$$

$$= 181259.031 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 181259 + 6713(103 - 9)^2$$

$$= 59248560.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	373.0	18.0	9.0	6713.3	60419.7	0.592E+08
Wearplate	180.0	6.0	21.0	1080.0	22680.0	0.723E+07
Web	10.0	290.0	169.0	2900.0	490099.8	0.330E+08
BasePlate	150.0	16.0	322.0	2400.0	772799.8	0.115E+09
Totals	...	...	...	13093.3	1345999.2	0.215E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1345999 / 13093$$

$$= 102.801 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 181700$$

$$= 36979.7852 \text{ N}$$



Tension Stress, $St = (Fh/As) = 5.7967 \text{ MPa}$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 155.9972 \text{ MPa}$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians})$
 $= 1180 - 850 \cdot \sin(120/2)/1.05$
 $= 477.056 \text{ mm}$

Bending Moment, $M = Fh * d = 17648564.0000 \text{ N-mm}$

Bending Stress, $Sb = (M * C1 / I) = 8.4406 \text{ MPa}$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 173.3302 \text{ MPa}$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 * \text{BasePlateLength} * \text{AllStress})}$
 $= \sqrt{3(181700 + 1444)150 / (2 * 1575 * 173)}$
 $= 12.286 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2)(\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2)(850 + 18 + 6)$
 $= 1513.812 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1514 / (3 - 1)$
 $= 756.906 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 757 \cdot 150/2$
 $= 56767.965 \text{ mm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 183143 / (1575 \cdot 150)$
 $= 0.775 \text{ N/mm}^2$

Axial Load [P]:
 $= Ap \cdot Bp$
 $= 56768 \cdot 0.78$
 $= 44007.117 \text{ N}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 1150 + 3785$
 $= 4934.531 \text{ mm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 44007/4935$
 $= 8.919 \text{ MPa}$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	115.0	67.5	1150.0	77625.0	0.391E+07
Web	378.5	10.0	5.0	3784.5	18922.7	0.834E+06
Totals	...	...	...	4934.5	96547.6	0.474E+07

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 125 - 10$
 $= 115.000 \text{ mm}$



Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 96548/4935$$

$$= 19.566 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{Saddle Width} - ytot)$$

$$= \max(19.6, 125 - 19.6)$$

$$= 105.434 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 4744205/4935 }$$

$$= 31.007 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1180 - \cos(120/2) (850 + 18 + 6) - 16$$

$$= 727.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$$

$$= \sqrt{ 2 \cdot \pi^2 \cdot 199955/260 }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 727/31$$

$$= 23.446$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 4453 / (2 \cdot 1575) \cdot 757 \cdot 727/2$$

$$= 389094.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($23.4 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.4)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (23.4) / (8 \cdot 123) - (23.4^3) / (8 \cdot 123^3))$$

$$= 147 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 8.92/147 + (389094 \cdot 105/4744210) / 173$$

$$= 0.111$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	115.0	67.5	1150.0	77625.0	0.465E+07
Web	756.9	10.0	5.0	7569.1	37845.3	0.577E+06
Totals	...	...	...	8719.1	115470.3	0.523E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 115470/8719$$

$$= 13.243 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(13.2, \text{abs}(125 - 13.2))$$

$$= 111.757 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1180 - \cos(874/2) (16 + 0 + 0) - 0$$

$$= 290.000 \text{ mm}$$



Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{5230157/8719}$$

$$= 24.492 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 290/24.5$$

$$= 11.841$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 4453 / (2 \cdot 1575)$$

$$= 1.414 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.41 \cdot 757 \cdot 290$$

$$= 310418.812 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 310419/46800$$

$$= 6.631 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($11.8 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.8)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (11.8) / (8 \cdot 123) - (11.8)^3 / (8 \cdot 123^3))$$

$$= 152 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 10.1/152 + (310419 \cdot 112/5230157) / 173$$

$$= 0.105$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	225.7742 mm ²
Saddle Load Q0 (Weight)	Q0	160653.8 N
Saddle Load QL (Wind/Seismic contribution)	QL	1083.4 N
Maximum Transverse Force	Ft	9617.3 N
Maximum Longitudinal Force	Fl	4452.9 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4453 / (226 \cdot 4)$$

$$= 4.9 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 9617 / (226 \cdot 4)$$

$$= 10.7 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$



Bolt Area due to Shear Load [Bltarears]:
 $= F / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 4453 / (103 \cdot 4)$
 $= 10.7882 \text{ mm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1180 \cdot 9617 + 0$
 $= 11352976.00 \text{ N-mm}$

Eccentricity (e):
 $= \text{Rmom} / \text{QO}$
 $= 11352976 / 160654$
 $= 70.64 \text{ mm} < \text{Bplen} / 6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0, 10.8, 0]$
 $= 10.7882 \text{ mm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{\text{rm} \cdot \text{t}}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{ \text{Rm} \cdot \text{t} }, 2a)$
 $= \min(125 + 1.56 \cdot \sqrt{ 859 \cdot 18 }, 2 \cdot 1550)$
 $= 318.9802 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	859.00	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	7950.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1550.00	mm
Saddle Width	b1 or b	125.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	434.00	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1180.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 214092.89 N
 Pressure used in Analysis P 3.411 MPa

Horizontal Vessel Analysis Results:		Actual MPa	Allowable MPa
Long. Stress at Top of Midspan		79.67	234.00
Long. Stress at Bottom of Midspan		83.11	234.00
Long. Stress at Top of Saddles		98.46	234.00
Long. Stress at Bottom of Saddles		71.93	234.00



Tangential Shear in Shell	9.22	138.58
Circ. Stress at Horn of Saddle	61.71	350.99
Circ. Compressive Stress in Shell	2.84	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (19235/2 + 0) \cdot 1180/1514$$

$$= 22489.7 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(4453, 0) \cdot 1180/4850$$

$$= 1083.4 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 191603 + \max(1083, 22490, 0, 0)$$

$$= 214092.9 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1844 (2.5 + \pi/4 (1.92 \cdot 1.2)^2)$$

$$= 13494.322 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	215536.84	N
Transverse Shear Load	9617.27	N
Longitudinal Shear Load	4452.90	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1700/2 + 18 + 6) \sin(120/2), 2 \cdot 859)$$

$$= 1513.812 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -214093 \cdot 1550 [1 - (1 - 1550/7950 + (859^2 - 434^2)/(2 \cdot 1550 \cdot 7950)) / (1 + (4 \cdot 434)/(3 \cdot 7950))]$$

$$= -75958304.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 214093 \cdot 7950 / 4 ((1 + 2(859^2 - 434^2)/(7950^2)) / (1 + (4 \cdot 434)/(3 \cdot 7950)) - 4 \cdot 1550/7950)$$

$$= 71721296.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 3.41 \cdot 859 / (2 \cdot 18) - 71721296 / (\pi \cdot 859^2 \cdot 18)$$

$$= 79.67 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 3.41 \cdot 859 / (2 \cdot 18) + 71721296 / (\pi \cdot 859^2 \cdot 18)$$

$$= 83.11 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$



$$= 3.41 \cdot 859 / (2 \cdot 18) - 75958304 / (0.11 \cdot \pi \cdot 859^2 \cdot 18)$$

$$= 98.46 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 3.41 \cdot 859 / (2 \cdot 18) + 75958304 / (0.19 \cdot \pi \cdot 859^2 \cdot 18)$$

$$= 71.93 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 214093 (7950 - 2 \cdot 1550) / (7950 + (4 \cdot 434 / 3))$$

$$= 121748.3 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 121748 / (859 \cdot 18)$$

$$= 9.22 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(859 \cdot 18)}$$

$$= 96.990 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 214093 \cdot 0.1 / (18(125 + 97 + 97))$$

$$= -2.84 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -214093 / (4 \cdot 18(125 + 97 + 97)) - 3 \cdot 0.053 \cdot 214093 / (2 \cdot 18^2)$$

$$= -61.71 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(125 + 1.56 \cdot \sqrt{(859 \cdot 18)}, 2 \cdot 1550)$$

$$= 318.98 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1575.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	290.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(850 \cdot 18)} + 180)^3) / 12$$

$$= 181259.031 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 181259 + 6713(103 - 9)^2$$

$$= 59248560.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	373.0	18.0	9.0	6713.3	60419.7	0.592E+08
Wearplate	180.0	6.0	21.0	1080.0	22680.0	0.723E+07
Web	10.0	290.0	169.0	2900.0	490099.8	0.330E+08



BasePlate	150.0	16.0	322.0	2400.0	772799.8	0.115E+09
Totals	...	...	...	13093.3	1345999.2	0.215E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1345999/13093$$

$$= 102.801 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 214093$$

$$= 43572.5469 \text{ N}$$

$$\text{Tension Stress, } St = (Fh/As) = 6.8301 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1180 - 850 \cdot \sin(120/2) / 1.05$$

$$= 477.056 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 20794952.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 9.9454 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(214093 + 1444) \cdot 150) / (2 \cdot 1575 \cdot 173)}$$

$$= 13.329 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (850 + 18 + 6)$$

$$= 1513.812 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 1514 / (3 - 1)$$

$$= 756.906 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 757 \cdot 150/2$$

$$= 56767.965 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 215537 / (1575 \cdot 150)$$

$$= 0.912 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 56768 \cdot 0.91$$

$$= 51790.852 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



$$= 1150 + 3785$$

$$= 4934.531 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 51791/4935$$

$$= 10.496 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	115.0	67.5	1150.0	77625.0	0.391E+07
Web	378.5	10.0	5.0	3784.5	18922.7	0.834E+06
Totals	...	...	...	4934.5	96547.6	0.474E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 125 - 10$$

$$= 115.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 96548/4935$$

$$= 19.566 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(19.6, 125 - 19.6)$$

$$= 105.434 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 4744205/4935 }$$

$$= 31.007 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1180 - \cos(120/2) (850 + 18 + 6) - 16$$

$$= 727.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} }$$

$$= \sqrt{ 2 \cdot \pi^2 \cdot 199955/260 }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 727/31$$

$$= 23.446$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 4453 / (2 \cdot 1575) \cdot 757 \cdot 727/2$$

$$= 389094.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (23.4 < 123) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (23.4)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (23.4) / (8 \cdot 123) - (23.4)^3 / (8 \cdot 123^3))$$

$$= 147 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 10.5/147 + (389094 \cdot 105/4744210) / 173$$

$$= 0.121$$

Check of Inside Ribs:



Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	115.0	67.5	1150.0	77625.0	0.465E+07
Web	756.9	10.0	5.0	7569.1	37845.3	0.577E+06
Totals	...	...	...	8719.1	115470.3	0.523E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 115470/8719$$

$$= 13.243 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(13.2, \text{abs}(125 - 13.2))$$

$$= 111.757 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1180 - \cos(874/2) (16 + 0 + 0) - 0$$

$$= 290.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (5230157/8719) }$$

$$= 24.492 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 290/24.5$$

$$= 11.841$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4453/(2 \cdot 1575)$$

$$= 1.414 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.41 \cdot 757 \cdot 290$$

$$= 310418.812 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 310419/46800$$

$$= 6.631 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($11.8 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.8)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (11.8) / (8 \cdot 123) - (11.8^3) / (8 \cdot 123^3))$$

$$= 152 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 11.9/152 + (310419 \cdot 112/5230157) / 173$$

$$= 0.116$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification	SA-193 B7	
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa



Area Available in a Single Bolt	BltArea	225.7742 mm ²
Saddle Load QO (Weight)	QO	193047.2 N
Saddle Load QL (Wind/Seismic contribution)	QL	1083.4 N
Maximum Transverse Force	Ft	9617.3 N
Maximum Longitudinal Force	Fl	4452.9 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $4453 / (226 \cdot 4)$
= 4.9 MPa. Must be less than 103.2 MPa.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $9617 / (226 \cdot 4)$
= 10.7 MPa. Must be less than 103.2 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $4453 / (103 \cdot 4)$
= 10.7882 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1180 \cdot 9617 + 0$
= 11352976.00 N-mm

Eccentricity (e):
= R_{mom} / QO
= $11352976 / 193047$
= 58.79 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0, 10.8, 0]$
= 10.7882 mm²

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(125 + 1.56 \cdot \sqrt{ 860 \cdot 16.5 }, 2 \cdot 1550)$$

$$= 310.8030 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	859.75	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	7950.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1550.00	mm
Saddle Width	b1 or b	125.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	434.75	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1180.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	197201.14	N
Pressure used in Analysis	P	-0.103 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	-4.42	-218.70
Long. Stress at Top of Midspan	4.42	138.00
Long. Stress at Bottom of Midspan	-0.97	-218.70
Long. Stress at Bottom of Midspan	0.97	138.00
Long. Stress at Top of Saddles	14.43	138.00
Long. Stress at Bottom of Saddles	-12.19	-218.70
Long. Stress at Bottom of Saddles	12.19	138.00
Tangential Shear in Shell	9.25	110.40
Circ. Stress at Horn of Saddle	67.04	172.50
Circ. Compressive Stress in Shell	2.92	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (58286/2 + 0) \cdot 1180/1514$$

$$= 68150.5 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(13494, 0) \cdot 1180/4850$$

$$= 3283.0 \text{ N}$$

Saddle Reaction Force due to Earthquake FI [Fsl]:



$$= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(25767, 0) \cdot 1180/4850$$

$$= 6269.1 \text{ N}$$

Saddle Reaction Force due to Earthquake F_t [Fst]:

$$= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (25767/2 + 0) \cdot 1180/1514$$

$$= 30127.5 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 129051 + \max(3283, 68150, 6269, 30128)$$

$$= 197201.1 \text{ N}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1844 (2.5 + \pi/4(1.92 \cdot 1.2)^2)$$

$$= 13494.322 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	198645.09	N
Transverse Shear Load	29143.23	N
Longitudinal Shear Load	25766.92	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1700/2 + 18 + 6) \sin(120/2), 2 \cdot 860)$$

$$= 1513.812 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/(3L))]$$

$$= -197201 \cdot 1550 [1 - (1 - 1550/7950 + (860^2 - 435^2)/(2 \cdot 1550 \cdot 7950)) / (1 + (4 \cdot 435)/(3 \cdot 7950))]$$

$$= -69985528.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/(3L)) - 4a/L)$$

$$= 197201 \cdot 7950/4 ((1 + 2(860^2 - 435^2)/(7950^2)) / (1 + (4 \cdot 435)/(3 \cdot 7950)) - 4 \cdot 1550/7950)$$

$$= 66026364.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= -0.1 \cdot 860/(2 \cdot 16.5) - 66026364/(\pi \cdot 860^2 \cdot 16.5)$$

$$= -4.42 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= -0.1 \cdot 860/(2 \cdot 16.5) + 66026364/(\pi \cdot 860^2 \cdot 16.5)$$

$$= -0.97 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$$

$$= -0.1 \cdot 860/(2 \cdot 16.5) - 69985528/(0.11 \cdot \pi \cdot 860^2 \cdot 16.5)$$

$$= 14.43 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M_1/(K_1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.1 \cdot 860/(2 \cdot 16.5) + 69985528/(0.19 \cdot \pi \cdot 860^2 \cdot 16.5)$$

$$= -12.19 \text{ MPa}$$



Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a)/(L+(4 \cdot hm/3))$
 $= 197201 (7950 - 2 \cdot 1550)/(7950 + (4 \cdot 435/3))$
 $= 112129.3 \text{ N}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K_2 \cdot T / (R_m \cdot t)$
 $= 1.17 \cdot 112129/(860 \cdot 16.5)$
 $= 9.25 \text{ MPa}$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(860 \cdot 16.5)}$
 $= 92.901 \text{ mm}$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K_5 \cdot Q \cdot k / (t (b + X_1 + X_2))$
 $= -0.76 \cdot 197201 \cdot 0.1/(16.5 (125 + 92.9 + 92.9))$
 $= -2.92 \text{ MPa}$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:
 $= -Q/(4 \cdot t(b+X_1+X_2)) - 3 \cdot K_7 \cdot Q/(2 \cdot t^2)$
 $= -197201/(4 \cdot 16.5 (125 + 92.9 + 92.9)) -$
 $3 \cdot 0.053 \cdot 197201/(2 \cdot 16.5^2)$
 $= -67.04 \text{ MPa}$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$
 $= \min(125 + 1.56 \cdot \sqrt{(860 \cdot 16.5)}, 2 \cdot 1550)$
 $= 310.80 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1575.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	243.0	MPa
Height of Web at Center	Hw,c	290.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(852 \cdot 16.5)} + 180)16.5^3)/12$
 $= 136601.438 \text{ mm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 136601 + 6021 (107 - 8.25)^2$
 $= 58733384.000 \text{ mm}^4$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	364.9	16.5	8.3	6021.0	49673.3	0.587E+08
Wearplate	180.0	6.0	19.5	1080.0	21060.0	0.825E+07
Web	10.0	290.0	167.5	2900.0	485749.8	0.310E+08
BasePlate	150.0	16.0	320.5	2400.0	769199.8	0.110E+09
Totals	...	...	...	12401.0	1325682.8	0.208E+09

Distance to Centroid [C1]:
 $= AY / A$
 $= 1325682/12401$
 $= 106.901 \text{ mm}$



Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 197201 \\
 &= 40134.7070 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 6.2912 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 145.7982 \text{ MPa}
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1180 - 852 \cdot \sin(120/2) / 1.05 \\
 &= 475.815 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 19104440.0000 \text{ N-mm}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 9.8387 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 161.9980 \text{ MPa}
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(197201 + 1444)150 / (2 \cdot 1575 \cdot 162)} \\
 &= 13.236 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (850 + 18 + 6) \\
 &= 1513.812 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 1514 / (3 - 1) \\
 &= 756.906 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 757 \cdot 150 / 2 \\
 &= 56767.965 \text{ mm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 198645 / (1575 \cdot 150) \\
 &= 0.841 \text{ N/mm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 56768 \cdot 0.84 \\
 &= 47731.969 \text{ N}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 1150 + 3785 \\
 &= 4934.531 \text{ mm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P/Ar \\
 &= 47732/4935 \\
 &= 9.674 \text{ MPa}
 \end{aligned}$$



Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	115.0	67.5	1150.0	77625.0	0.391E+07
Web	378.5	10.0	5.0	3784.5	18922.7	0.834E+06
Totals	...	...	...	4934.5	96547.6	0.474E+07

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 125 - 10
 = 115.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 96548/4935
 = 19.566 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(19.6, 125 - 19.6)
 = 105.434 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4744205/4935}$
 = 31.007 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1180 - cos(120/2)(850 + 18 + 6) - 16
 = 727.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199955/243}$
 = 127.447

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 727/31
 = 23.446

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 25767 / (2 • 1575) • 757 • 727/2
 = 2251513.500 N-mm

Compressive Allowable, KL/r < Cc (23.4 < 127) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (23.4)^2 / (2 \cdot 127^2)) 243 / (5/3 + 3 \cdot (23.4) / (8 \cdot 127) - (23.4)^3 / (8 \cdot 127^3))$
 = 137.7 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 9.67/138 + (2251513 • 105/4744210) / 162
 = 0.379

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	115.0	67.5	1150.0	77625.0	0.465E+07
Web	756.9	10.0	5.0	7569.1	37845.3	0.577E+06
Totals	...	...	...	8719.1	115470.3	0.523E+07



Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 115470/8719$$

$$= 13.243 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(13.2, \text{abs}(125 - 13.2))$$

$$= 111.757 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1180 - \cos(874/2)(16 + 0 + 0) - 0$$

$$= 290.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(5230157/8719)}$$

$$= 24.492 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 290/24.5$$

$$= 11.841$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 25767/(2 \cdot 1575)$$

$$= 8.180 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 8.18 \cdot 757 \cdot 290$$

$$= 1796255.375 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1796255/46800$$

$$= 38.370 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (11.8 < 127) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.8)^2 / (2 \cdot 127^2)) 243 / (5/3 + 3 \cdot (11.8) / (8 \cdot 127) - (11.8^3) / (8 \cdot 127^3))$$

$$= 142.2 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 11/142 + (1796255 \cdot 112/5230157) / 162$$

$$= 0.314$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(125 + 1.56 \cdot \sqrt{ 860 \cdot 16.5 }, 2 \cdot 1550)$$

$$= 310.8030 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	859.75	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	7950.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1550.00	mm
Saddle Width	b1 or b	125.00	mm



Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	434.75	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1180.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		231013.77	N
Pressure used in Analysis	P	-0.103	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	-4.71	-218.70
Long. Stress at Top of Midspan	4.71	138.00
Long. Stress at Bottom of Midspan	-0.68	-218.70
Long. Stress at Bottom of Midspan	0.68	138.00
Long. Stress at Top of Saddles	17.37	138.00
Long. Stress at Bottom of Saddles	-13.82	-218.70
Long. Stress at Bottom of Saddles	13.82	138.00
Tangential Shear in Shell	10.84	110.40
Circ. Stress at Horn of Saddle	78.54	172.50
Circ. Compressive Stress in Shell	3.43	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (58286/2 + 0) \cdot 1180/1514$
 $= 68150.5 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(13494, 0) \cdot 1180/4850$
 $= 3283.0 \text{ N}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(25767, 0) \cdot 1180/4850$
 $= 6269.1 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (25767/2 + 0) \cdot 1180/1514$
 $= 30127.5 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 162863 + \max(3283, 68150, 6269, 30128)$
 $= 231013.8 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1844 (2.5 + \pi/4 (1.92 \cdot 1.2)^2)$
 $= 13494.322 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	232457.70	N
Transverse Shear Load	29143.23	N
Longitudinal Shear Load	25766.92	N



Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1700/2 + 18 + 6)\sin(120/2), 2 \cdot 860)$$

$$= 1513.812 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -231014 \cdot 1550 [1 - (1 - 1550/7950 + (860^2 - 435^2)/(2 \cdot 1550 \cdot 7950))/(1 + (4 \cdot 435)/(3 \cdot 7950))]$$

$$= -81985432.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 231014 \cdot 7950/4((1 + 2(860^2 - 435^2)/(7950^2))/(1 + (4 \cdot 435)/(3 \cdot 7950)) - 4 \cdot 1550/7950)$$

$$= 77347424.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= -0.1 \cdot 860/(2 \cdot 16.5) - 77347424/(\pi \cdot 860^2 \cdot 16.5)$$

$$= -4.71 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= -0.1 \cdot 860/(2 \cdot 16.5) + 77347424/(\pi \cdot 860^2 \cdot 16.5)$$

$$= -0.68 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= -0.1 \cdot 860/(2 \cdot 16.5) - 81985432/(0.11 \cdot \pi \cdot 860^2 \cdot 16.5)$$

$$= 17.37 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.1 \cdot 860/(2 \cdot 16.5) + 81985432/(0.19 \cdot \pi \cdot 860^2 \cdot 16.5)$$

$$= -13.82 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 231014 (7950 - 2 \cdot 1550)/(7950 + (4 \cdot 435/3))$$

$$= 131355.3 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 131355/(860 \cdot 16.5)$$

$$= 10.84 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{860 \cdot 16.5}$$

$$= 92.901 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 231014 \cdot 0.1/(16.5(125 + 92.9 + 92.9))$$

$$= -3.43 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q/(4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q/(2 \cdot t^2)$$

$$= -231014/(4 \cdot 16.5(125 + 92.9 + 92.9)) - 3 \cdot 0.053 \cdot 231014/(2 \cdot 16.5^2)$$



= -78.54 MPa

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(125 + 1.56 \cdot \sqrt{(860 \cdot 16.5)}, 2 \cdot 1550)$$

$$= 310.80 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1575.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	243.0	MPa
Height of Web at Center	Hw,c	290.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(852 \cdot 16.5)} + 180) 16.5^3) / 12$$

$$= 136601.438 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 136601 + 6021 (107 - 8.25)^2$$

$$= 58733384.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	364.9	16.5	8.3	6021.0	49673.3	0.587E+08
Wearplate	180.0	6.0	19.5	1080.0	21060.0	0.825E+07
Web	10.0	290.0	167.5	2900.0	485749.8	0.310E+08
BasePlate	150.0	16.0	320.5	2400.0	769199.8	0.110E+09
Totals	...	...	...	12401.0	1325682.8	0.208E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1325682 / 12401$$

$$= 106.901 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 231014$$

$$= 47016.3125 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 7.3700 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 145.7982 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1180 - 852 \cdot \sin(120 / 2) / 1.05$$

$$= 475.815 \text{ mm}$$



Bending Moment, $M = F_h \cdot d = 22380138.0000 \text{ N-mm}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 11.5257 \text{ MPa}$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 161.9980 \text{ MPa}$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt{3(231014 + 1444)150 / (2 \cdot 1575 \cdot 162)}$
 $= 14.318 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (850 + 18 + 6)$
 $= 1513.812 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1514 / (3 - 1)$
 $= 756.906 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 757 \cdot 150 / 2$
 $= 56767.965 \text{ mm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 232458 / (1575 \cdot 150)$
 $= 0.984 \text{ N/mm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 56768 \cdot 0.98$
 $= 55856.727 \text{ N}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 1150 + 3785$
 $= 4934.531 \text{ mm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 55857 / 4935$
 $= 11.321 \text{ MPa}$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	10.0	115.0	67.5	1150.0	77625.0	0.391E+07
Web	378.5	10.0	5.0	3784.5	18922.7	0.834E+06
Totals	...	...	...	4934.5	96547.6	0.474E+07

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 125 - 10$
 $= 115.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 96548 / 4935$
 $= 19.566 \text{ mm}$

Distance to Centroid [C1]:
 $= \max(y_{tot}, \text{Saddle Width} - y_{tot})$
 $= \max(19.6, 125 - 19.6)$



$$= 105.434 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{4744205/4935}$$

$$= 31.007 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1180 - \cos(120/2)(850 + 18 + 6) - 16$$

$$= 727.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/243)}$$

$$= 127.447$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 727/31$$

$$= 23.446$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 25767 / (2 \cdot 1575) \cdot 757 \cdot 727/2$$

$$= 2251513.500 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($23.4 < 127$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (23.4)^2 / (2 \cdot 127^2)) 243 /$$

$$(5/3 + 3 \cdot (23.4) / (8 \cdot 127) - (23.4^3) / (8 \cdot 127^3))$$

$$= 137.7 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.3/138 + (2251513 \cdot 105/4744210) / 162$$

$$= 0.391$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	115.0	67.5	1150.0	77625.0	0.465E+07
Web	756.9	10.0	5.0	7569.1	37845.3	0.577E+06
Totals	...	...	...	8719.1	115470.3	0.523E+07

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 115470/8719$$

$$= 13.243 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{abs}(\text{Saddle Width} - \text{ytot}))$$

$$= \max(13.2, \text{abs}(125 - 13.2))$$

$$= 111.757 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1180 - \cos(874/2)(16 + 0 + 0) - 0$$

$$= 290.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{5230157/8719}$$

$$= 24.492 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 290/24.5$$



$$= 11.841$$

Unit Force [Force,u]:
= $F_l / (2 \cdot \text{Baseplate Length})$
= $25767 / (2 \cdot 1575)$
= 8.180 N/mm

Moment at base of inner Rib [Mbase,c]:
= Unit Force $\cdot e \cdot L$
= $8.18 \cdot 757 \cdot 290$
= 1796255.375 N-mm

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
= Bending Moment / Section Modulus
= $1796255 / 46800$
= 38.370 MPa

Compressive Allowable, $KL/r < C_c$ ($11.8 < 127$) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
= $(1 - (11.8)^2 / (2 \cdot 127^2)) 243 / (5/3 + 3 \cdot (11.8) / (8 \cdot 127) - (11.8^3) / (8 \cdot 127^3))$
= 142.2 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)
= $S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
= $12.8 / 142 + (1796255 \cdot 112 / 5230157) / 162$
= 0.327

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: MH-124"

From: 10

Pressure for Reinforcement Calculations	P	2.6054	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	1700.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	18.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-350 LF2 +IT	
-----------------	--	----------------	--

Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	88.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	118.3760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH-124"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		24.000	in.
Actual Thickness Used in Calculation		0.472	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]



$$\begin{aligned} &= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)} \\ &= 2.61 \cdot 1703 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.61) \\ &= 14.4718 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 2.61 \cdot 306 / (138 \cdot 1 - 0.6 \cdot 2.61) \\ &= 5.8492 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.0036 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$\begin{aligned} &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\ &= 88 / 0.5773 \\ &= 152.4205 \text{ mm} \end{aligned}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.2000	mm
Parallel to Vessel Wall, opening length	d	612.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Weld Strength Reduction Factor [fr1]:

$$\begin{aligned} &= \min(1, S_n / S_v) \\ &= \min(1, 138 / 138) \\ &= 1.000 \end{aligned}$$

Weld Strength Reduction Factor [fr2]:

$$\begin{aligned} &= \min(1, S_n / S_v) \\ &= \min(1, 138 / 138) \\ &= 1.000 \end{aligned}$$

Weld Strength Reduction Factor [fr3]:

$$\begin{aligned} &= \min(fr2, fr4) \\ &= \min(1, 1) \\ &= 1.000 \end{aligned}$$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8865.413	1376.778	NA
Area in Shell	A1	1242.487	7354.345	NA
Area in Nozzle Wall	A2	383.688	783.456	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	7260.000	7260.000	NA
TOTAL AREA AVAILABLE	Atot	8986.176	15497.801	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned} &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\ &= (613 \cdot 14.5 \cdot 1 + 2 \cdot 98.5 \cdot 14.5 \cdot 1 \cdot (1 - 1)) \\ &= 8865.413 \text{ mm}^2 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1



Area Available in Shell [A1]:

$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 613(1 \cdot 16.5 - 1 \cdot 14.5) - 2 \cdot 98.5 \\ &\quad (1 \cdot 16.5 - 1 \cdot 14.5) \cdot (1 - 1) \\ &= 1242.487 \text{ mm}^2 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot tlnp)(tn - trn)fr2 \\ &= (2 \cdot 41.2)(10.5 - 5.85)1 \\ &= 383.688 \text{ mm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\ &= (2 \cdot \min(41.2, 118, 60)) \cdot (100 - 12) \cdot 1 \\ &= 7260.000 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 7.3492 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= ta \\ &= 7.3492 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 18$, $tr = 14.5$, $c = 1.5 \text{ mm}$, $E* = 1$
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.88$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-17 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-77 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 2.61/5.10 = 0.511$$

Weld Size Calculations, Description: MH-124"

Intermediate Calc. for nozzle/shell Welds $Tmin$ 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot Wo$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$$



$$= \max(0, (8865 - 1242 + 2 \cdot 98.5 \cdot 1 \cdot (1 \cdot 16.5 - 14.5)) \cdot 138) \\ = 1106988.88 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot \text{Sv} \\ = (7644 + 100 - 0 \cdot 1) \cdot 138 \\ = 1068518.75 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr}1)) \cdot \text{Sv} \\ = (7644 + 0 + 100 + (3251)) \cdot 138 \\ = 1517041.50 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr}1)) \cdot \text{S} \\ = (7644 + 0 + 100 + 0 + (3251)) \cdot 138 \\ = 1517041.50 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wo} \cdot 0.49 \cdot \text{Snw} \\ = (3.14/2.0) \cdot 810 \cdot 10 \cdot 0.49 \cdot 138 \\ = 859846. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (\text{Dlr} + \text{Dlo}) / 4) \cdot (\text{Thk} - \text{Can}) \cdot 0.7 \cdot \text{Sn} \\ = (3.14 \cdot 356) \cdot (100 - 1.5) \cdot 0.7 \cdot 138 \\ = 10627207. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot (\text{T} - \text{cas}) \cdot 0.74 \cdot \text{Sng} \\ = (3.14/2.0) \cdot 810 \cdot (18 - 1.5) \cdot 0.74 \cdot 138 \\ = 2142596. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (859846 + 10627207) = 11487053 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (859846 + 0 + 2142596 + 0) = 3002442 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (859846 + 2142596 + 0) = 3002442 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 11487053 N	, must exceed W = 1106988 N	or W1 = 1068518 N
Path 2-2 = 3002442 N	, must exceed W = 1106988 N	or W2 = 1517041 N
Path 3-3 = 3002442 N	, must exceed W = 1106988 N	or W3 = 1517041 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.62 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 53.9946 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 190.3705 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N18"

From: 20

Pressure for Reinforcement Calculations	P	2.6002	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		310.00	mm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-350 LF2 +IT	
-----------------	--	----------------	--

Hub Height of Integral Nozzle	h	40.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		54.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	332.2720	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N18"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		7.625	in.
Actual Thickness Used in Calculation		0.500	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)



$$= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 16.2274 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.6 \cdot 98.3 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 1.8741 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.8356 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 41.3 / 0.5773$$

$$= 71.5337 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	393.3500	mm
Parallel to Vessel Wall, opening length	d	196.6750	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3191.529	1051.445	NA
Area in Shell	A1	53.609	1142.247	NA
Area in Nozzle Wall	A2	769.390	855.062	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	3405.716	3405.716	NA
TOTAL AREA AVAILABLE	Atot	4328.714	5503.025	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (197 \cdot 16.2 \cdot 1 + 2 \cdot 52.5 \cdot 16.2 \cdot 1 \cdot (1 - 1))$$

$$= 3191.529 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:



$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 197(1 \cdot 16.5 - 1 \cdot 16.2) - 2 \cdot 52.5$$

$$(1 \cdot 16.5 - 1 \cdot 16.2) \cdot (1 - 1)$$

$$= 53.609 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 41.2)(11.2 - 1.87)1$$

$$= 769.390 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(41.2, 332, 40)) \cdot (54 - 12.7) \cdot 1$$

$$= 3405.716 \text{ mm}^2 \text{ Includes Bevel Area of } 101.716 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3741 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.7274 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.7274 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.1385 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 8.6600 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.66, \max(17.7, 3)]$$

$$= 8.6600 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.37, 8.66)$$

$$= 8.6600 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 32.2,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 312.8 MPa	Passed
Occasional	: 10.8,	Allowable	: 183.5 MPa	Passed
Shear	: 15.0,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C

Flange MDMT with Temp reduction per UCS-66(i)(2) -77 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = 2.60/5.10 = 0.510

Weld Size Calculations, Description: N18"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (3192 - 53.6 + 2 • 52.5 • 1 •
(1 • 16.5 - 16.2))138)
= 436937.50 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (4175 + 100 - 0 • 1) • 138
= 589903.75 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (4175 + 0 + 100 + (1732)) • 138
= 828964.06 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (4175 + 0 + 100 + 0 + (1732)) • 138
= 828964.06 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 302 • 10 • 0.49 • 138
= 320398. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 125) • (54 - 1.5) • 0.7 • 138
= 1984796. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 302 • (18 - 1.5) • 0.74 • 138
= 798379. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (320398 + 1984796) = 2305194 N



Failure Path [Path 2-2]:
 = (Sonw + Tpgw + Tngw + Sinw)
 = (320398 + 0 + 798379 + 0) = 1118777 N

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (320398 + 798379 + 0) = 1118777 N

Summary of Failure Path Calculations:

Path 1-1 = 2305194 N , must exceed W = 436937 N or W1 = 589903 N
 Path 2-2 = 1118777 N , must exceed W = 436937 N or W2 = 828964 N
 Path 3-3 = 1118777 N , must exceed W = 436937 N or W3 = 828964 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 13.4906 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 363.7625 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 196.68 mm

Input Echo, WRC297 Item 1, Description: N18" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1700.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	193.675	mm
Nozzle Thickness	54.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2.60	MPa
Radial Load	P	-5400.01	N
Circumferential Shear	Vc	8000.00	N
Longitudinal Shear	Vl	8000.00	N
Circumferential Moment	Mc	0.480E+07	N-mm
Longitudinal Moment	Ml	0.640E+07	N-mm
Torsional Moment	Mt	0.800E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No



*Warning - The ratio, Dn/Tn (5.587) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 301.675 + 2 \cdot 1.65 \cdot \sqrt{(859.75(18.0 - 1.5))}$
 $= 694.720 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1719.500 mm
 Nozzle Outside Diameter (d) = 301.675 mm
 Vessel Thickness used (T) = 16.500 mm
 Nozzle Thickness used (t) = 52.500 mm
 $T / t = 0.314$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.791$$

Nr/P = 0.074
 Mr/P = 0.156
 M0/P = 0.045
 N0/P = 0.092
 MrD/Mc = 0.291
 NrDL/Mc = 0.092
 M0d/Mc = 0.090
 N0DL/Mc = 0.070
 MrD/Ml = 0.122
 NrDL/Ml = 0.083
 M0D/Ml = 0.039
 N0DL/Ml = 0.115

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	132	135	132	135
Outplane Stress Summary	113	141	167	122

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	66	66	66	66
Inplane Stress Summary	22	100	150	-1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------



Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	113	141	167	124

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-31	31	31	-31
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	66	66	66	66
Outplane Stress Summary	37	90	109	35

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-102	102	102	-102
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	132	135	132	135
Inplane Stress Summary	45	215	260	21

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	46	215	260	36

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0



Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7
Hoop Stress Summary	-2	0	15	18

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-2	1	10	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	4	3	15	18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7
Hoop Stress Summary	2	5	10	13

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-8	8	8	-8
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-6	7	14	-5

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	9	7	14	19

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	132	135	132	135	132	135	132	135
Circ. Pl (SUS)	-7	-7	9	9	-4	-4	6	6
Circ. Q (SUS)	-13	13	23	-23	-84	84	120	-120
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	-5	-5	7	7	-3	-3	5	5
Long. Q (SUS)	-39	39	75	-75	-26	26	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	132.0	135.0	132.0	135.0	132.0	135.0	132.0	135.0
Pm+Pl (SUS)	125.0	128.0	141.0	144.0	128.0	131.0	138.0	141.0
Pm+Pl+Q (Total)	112.2	141.4	164.2	123.1	44.5	215.0	258.1	36.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	135.00	138.00	Passed
Pm+Pl (SUS)	144.01	207.00	Passed
Pm+Pl+Q (TOTAL)	258.11	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	7	4	7	4	7	4	7
Circ. Pl (SUS)	-7	-7	9	9	-3	-3	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-2	2	4	-4	-7	7	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1

Pm (SUS)	4.0	7.0	4.0	7.0	4.0	7.0	4.0	7.0
Pm+Pl (SUS)	3.0	0.0	13.0	16.0	1.0	4.0	9.0	12.0
Pm+Pl+Q (Total)	3.6	2.8	13.2	16.1	8.2	7.3	13.2	17.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	138.00	Passed
Pm+Pl (SUS)	16.00	207.00	Passed
Pm+Pl+Q (TOTAL)	17.12	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N42"

From: 20

Pressure for Reinforcement Calculations	P	2.6107	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		200.00	mm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	241.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$
 $= 16.2938 \text{ mm}$



Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R(\exp([P/(\text{Sn} \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $26.9(\exp([2.61/(138 \cdot 1)] - 1))$
= 0.5137 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4206 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	117.3280	mm
Parallel to Vessel Wall	Rn+tn+t	58.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N42".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0137	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.7938	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.7938	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.1385	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(17.8, 3)]
= 6.3000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.01, 6.3)
= 6.3000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
(see 319.4.4 and 302.3.5):**

Sustained	:	11.6,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	333.4 MPa	Passed
Occasional	:	1.8,	Allowable	:	183.5 MPa	Passed
Shear	:	9.4,	Allowable	:	96.6 MPa	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
the expansion stress allowable.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.51, c = 1.5 mm, E* = 1
Thickness Ratio = $\text{tr} \cdot E^*/(\text{tg} - \text{c}) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N42"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640	mm
---	------	---------	----

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0464 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 260.4723 mm

Input Echo, WRC297 Item 1, Description: N42" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.5000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1700.000	mm	
Shell Thickness	18.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	1.5000	mm	
Nozzle Diameter	50.800	mm	
Nozzle Thickness	16.7640	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-350 LF2		
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	2.61	MPa
Radial Load	P	-1300.00	N
Circumferential Shear	Vc	2000.00	N
Longitudinal Shear	Vl	2000.00	N
Circumferential Moment	Mc	300000.00	N-mm
Longitudinal Moment	Ml	400000.03	N-mm
Torsional Moment	Mt	500000.00	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (4.685) is < 5
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (5.030) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(859.75(18.0 - 1.5))}$$

$$= 477.373 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1719.500 mm
Nozzle Outside Diameter	(d) =	84.328 mm
Vessel Thickness used	(T) =	16.500 mm
Nozzle Thickness used	(t) =	15.264 mm
T / t	=	1.081



$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.501$$

Nr/P	=	0.178
Mr/P	=	0.272
M0/P	=	0.100
N0/P	=	0.172
MrD/Mc	=	0.285
NrDL/Mc	=	0.148
M0d/Mc	=	0.122
N0DL/Mc	=	-0.101
MrD/Ml	=	0.257
NrDL/Ml	=	0.143
M0D/Ml	=	0.107
N0DL/Ml	=	-0.015

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-11	11	11	-11
Normal Pressure Stress	133	136	133	136
Outplane Stress Summary	126	145	148	122

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-26	26	26	-26
Inplane Pressure Stress	66	66	66	66
Inplane Stress Summary	46	84	104	35

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	126	145	148	122

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-9	9	9	-9
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	66	66	66	66

Outplane Stress Summary	62	75	78	53
-------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-22	22	22	-22
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	133	136	133	136

Inplane Stress Summary	117	149	166	108
------------------------	-----	-----	-----	-----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	117	149	166	108
------------------------	-----	-----	-----	-----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7

Hoop Stress Summary	5	8	4	7
---------------------	---	---	---	---

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-22	22	22	-22
Axial Pressure Stress	2	2	2	2



Axial Stress Summary -22 10 39 -18

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress) Au Al Bu Bl
 Outside Inside Outside Inside

Shear due to (Vc) 0 0 0 0
 Shear due to (Vl) 0 0 0 0
 Shear due to Torsion 2 2 2 2

Shear Stress Summary 3 3 1 1

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities) Au Al Bu Bl
 Outside Inside Outside Inside

Two * Max Shear Stress 28 13 39 26

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses in the hoop direction) Cu Cl Du Dl
 Left Left Right Right
 Outside Inside Outside Inside

Hoop Membrane (P) 0 0 0 0
 Hoop Bending (P) 0 0 0 0
 Hoop Membrane (Mc) 1 1 -1 -1
 Hoop Bending (Mc) 0 0 0 0
 Hoop Membrane (ML) 0 0 0 0
 Hoop Bending (ML) 0 0 0 0
 Hoop Pressure Stress 4 7 4 7

Hoop Stress Summary 6 9 3 6

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to pipe cross-section) Cu Cl Du Dl
 Left Left Right Right
 Outside Inside Outside Inside

Axial Membrane (P) 0 0 0 0
 Axial Bending (P) 6 -6 6 -6
 Axial Membrane (Mc) -6 -6 6 6
 Axial Bending (Mc) -19 19 19 -19
 Axial Membrane (ML) 0 0 0 0
 Axial Bending (ML) 0 0 0 0
 Axial Pressure Stress 2 2 2 2

Axial Stress Summary -16 9 34 -16

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress) Cu Cl Du Dl
 Outside Inside Outside Inside

Shear due to (Vc) 0 0 0 0
 Shear due to (Vl) 0 0 0 0
 Shear due to Torsion 2 2 2 2

Shear Stress Summary 1 1 3 3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities) Cu Cl Du Dl
 Outside Inside Outside Inside

Two * Max Shear Stress 23 11 34 24

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of	Stress Values at
---------	------------------

Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	133	136	133	136	133	136	133	136
Circ. Pl (SUS)	0	0	0	0	-1	-1	1	1
Circ. Q (SUS)	-9	9	13	-13	-15	15	29	-29
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	-2	-2	2	2	1	1	-1	-1
Long. Q (SUS)	-19	19	33	-33	-7	7	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	133.0	136.0	133.0	136.0	133.0	136.0	133.0	136.0
Pm+Pl (SUS)	133.0	136.0	133.0	136.0	132.0	135.0	134.0	137.0
Pm+Pl+Q (Total)	124.1	145.1	146.1	123.0	117.1	150.1	163.0	108.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+Pl (SUS)	137.00	207.00	Passed
Pm+Pl+Q (TOTAL)	163.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	7	4	7	4	7	4	7
Circ. Pl (SUS)	0	0	0	0	1	1	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-8	-8	8	8	-6	-6	6	6
Long. Q (SUS)	-16	16	28	-28	-13	13	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	4.0	7.0	4.0	7.0	4.0	7.0	4.0	7.0
Pm+Pl (SUS)	10.0	13.0	10.0	10.0	9.0	12.0	8.0	8.0
Pm+Pl+Q (Total)	26.3	11.0	38.1	25.3	22.4	10.6	33.1	23.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	138.00	Passed
Pm+Pl (SUS)	13.00	207.00	Passed
Pm+Pl+Q (TOTAL)	38.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 Licensee: SL User
FileName : V-31002_7%
Nozzle Calcs.: N4_2"

Page 114 of 218
May 20, 2026



HEXAGON

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K3A3"

From: 20

Pressure for Reinforcement Calculations	P	2.6024	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	500.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		2000.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		54.83	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K3A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.000	in.
Actual Thickness Used in Calculation		0.810	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 16.2415 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.6 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 0.7553 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	194.9944	mm
Parallel to Vessel Wall, opening length	d	97.4972	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	791.750	521.230	NA
Area in Shell	A1	816.954	566.243	NA
Area in Nozzle Wall	A2	1782.732	1810.427	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	2699.686	2476.670	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 54.32 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (97.5 \cdot 16.2 \cdot 0.5 + 2 \cdot 19.1 \cdot 16.2 \cdot 0.5 \cdot (1 - 1))$$

$$= 791.750 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 97.5 (1 \cdot 16.5 - 0.5 \cdot 16.2) - 2 \cdot 19.1$$

$$(1 \cdot 16.5 - 0.5 \cdot 16.2) \cdot (1 - 1)$$

$$= 816.954 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 41.2) (19.1 - 0.76) / \sin(58)$$

$$= 1782.732 \text{ mm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.2, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K3A3"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (792 - 817 + 2 • 19.1 • 1 •
(1 • 16.5 - 8.12))138)
= 40629.63 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1783 + 0 + 100 - 0 • 1) • 138
= 259790.25 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1783 + 0 + 100 + (629)) • 138
= 346644.31 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1783 + 0 + 100 + 0 + (629)) • 138
= 346644.31 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 144 • 10 • 0.49 • 138
= 153424. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 60.5) • (20.6 - 1.5) • 0.7 • 138
= 350106. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 144 • (18 - 1.5) • 0.74 • 138
= 382307. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (153424 + 350106) = 503530 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (153424 + 0 + 382307 + 0) = 535731 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (153424 + 382307 + 0) = 535731 N

Summary of Failure Path Calculations:



Path 1-1 = 503529 N , must exceed W = 40629 N or W1 = 259790 N
 Path 2-2 = 535731 N , must exceed W = 40629 N or W2 = 346644 N
 Path 3-3 = 535731 N , must exceed W = 40629 N or W3 = 346644 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$
 $= 16.2415 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.6 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.6)$
 $= 0.7553 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 158.4000 mm
 Parallel to Vessel Wall, opening length d 79.2000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 41.2500 mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1286.326	423.412	NA
Area in Shell	A1	20.474	459.977	NA
Area in Nozzle Wall	A2	1511.290	1534.768	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1631.764	2094.745	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (79.2 \cdot 16.2 \cdot 1 + 2 \cdot 19.1 \cdot 16.2 \cdot 1 \cdot (1 - 1))$
 $= 1286.326 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2 (1 \cdot 16.5 - 1 \cdot 16.2) - 2 \cdot 19.1$
 $(1 \cdot 16.5 - 1 \cdot 16.2) \cdot (1 - 1)$



$$= 20.474 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot t_{np})(t_n - t_r)fr_2 \\ &= (2 \cdot 41.2)(19.1 - 0.76)1 \\ &= 1511.290 \text{ mm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2553 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.7415 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.7415 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.1385 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 7.2300 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[7.23, \max(17.7, 3)] \\ &= 7.2300 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(2.26, 7.23) \\ &= 7.2300 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Weld Size Calculations, Description: K3A3"

Intermediate Calc. for nozzle/shell welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr))Sv) \\ &= \max(0, (1286 - 20.5 + 2 \cdot 19.1 \cdot 1 \cdot (1 \cdot 16.5 - 16.2))138) \\ &= 176030.36 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A5 + A4 - (W_i - Can/.707)^2 \cdot fr_2) \cdot Sv \\ &= (1511 + 0 + 100 - 0 \cdot 1) \cdot 138 \\ &= 222335.12 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv \\ &= (1511 + 0 + 100 + (629)) \cdot 138 \\ &= 309189.16 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (1511 + 0 + 100 + 0 + (629)) \cdot 138 \\ &= 309189.16 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$$



$$= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138$$
$$= 124631. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.14 \cdot 49.1) \cdot (20.6 - 1.5) \cdot 0.7 \cdot 138$$
$$= 284402. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$
$$= (3.14/2.0) \cdot 117 \cdot (18 - 1.5) \cdot 0.74 \cdot 138$$
$$= 310560. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (124631 + 284402) = 409033 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (124631 + 0 + 310560 + 0) = 435191 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$
$$= (124631 + 310560 + 0) = 435191 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 409032 N	, must exceed W = 176030 N	or W1 = 222335 N
Path 2-2 = 435191 N	, must exceed W = 176030 N	or W2 = 309189 N
Path 3-3 = 435191 N	, must exceed W = 176030 N	or W3 = 309189 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 46.7746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.9370 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K3B3"

From: 20

Pressure for Reinforcement Calculations	P	2.6084	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	500.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		2000.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		125.17	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K3B3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.000	in.
Actual Thickness Used in Calculation		0.810	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 16.2797 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.61 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 0.7571 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	194.9920	mm
Parallel to Vessel Wall, opening length	d	97.4960	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	793.604	521.224	NA
Area in Shell	A1	815.080	566.236	NA
Area in Nozzle Wall	A2	1782.559	1810.427	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	2697.639	2476.663	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 54.33 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (97.5 \cdot 16.3 \cdot 0.5 + 2 \cdot 19.1 \cdot 16.3 \cdot 0.5 \cdot (1 - 1))$$

$$= 793.604 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 97.5 (1 \cdot 16.5 - 0.5 \cdot 16.3) - 2 \cdot 19.1$$

$$(1 \cdot 16.5 - 0.5 \cdot 16.3) \cdot (1 - 1)$$

$$= 815.080 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 41.2) (19.1 - 0.76) / \sin(58)$$

$$= 1782.559 \text{ mm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.3, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.99, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K3B3"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (794 - 815 + 2 • 19.1 • 1 •
(1 • 16.5 - 8.14))138)
= 41043.37 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1783 + 0 + 100 - 0 • 1) • 138
= 259766.34 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1783 + 0 + 100 + (629)) • 138
= 346620.41 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1783 + 0 + 100 + 0 + (629)) • 138
= 346620.41 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 144 • 10 • 0.49 • 138
= 153422. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 60.5) • (20.6 - 1.5) • 0.7 • 138
= 350102. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 144 • (18 - 1.5) • 0.74 • 138
= 382302. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (153422 + 350102) = 503524 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (153422 + 0 + 382302 + 0) = 535724 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (153422 + 382302 + 0) = 535724 N

Summary of Failure Path Calculations:

Path 1-1 = 503523 N , must exceed W = 41043 N or W1 = 259766 N
 Path 2-2 = 535724 N , must exceed W = 41043 N or W2 = 346620 N
 Path 3-3 = 535724 N , must exceed W = 41043 N or W3 = 346620 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3B3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$
 $= 16.2797 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.61 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.61)$
 $= 0.7571 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 158.4000 mm
 Parallel to Vessel Wall, opening length d 79.2000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 41.2500 mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1289.354	423.412	NA
Area in Shell	A1	17.446	459.977	NA
Area in Nozzle Wall	A2	1511.144	1534.768	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1628.589	2094.745	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (79.2 \cdot 16.3 \cdot 1 + 2 \cdot 19.1 \cdot 16.3 \cdot 1 \cdot (1 - 1))$
 $= 1289.354 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2 (1 \cdot 16.5 - 1 \cdot 16.3) - 2 \cdot 19.1$
 $(1 \cdot 16.5 - 1 \cdot 16.3) \cdot (1 - 1)$



$$= 17.446 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot t_{np})(t_n - t_{rn})fr_2 \\ &= (2 \cdot 41.2)(19.1 - 0.76)1 \\ &= 1511.144 \text{ mm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.2571 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 17.7797 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 17.7797 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 2.1385 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb_3, \max(tb_1, tb_2)] \\ &= \min[7.23, \max(17.8, 3)] \\ &= 7.2300 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(2.26, 7.23) \\ &= 7.2300 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Weld Size Calculations, Description: K3B3"

Intermediate Calc. for nozzle/shell welds $T_{min} = 16.5000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v) \\ &= \max(0, (1289 - 17.4 + 2 \cdot 19.1 \cdot 1 \cdot (1 \cdot 16.5 - 16.3)) \cdot 138) \\ &= 176664.73 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_5 + A_4 - (W_i - Can/.707)^2 \cdot fr_2) \cdot S_v \\ &= (1511 + 0 + 100 - 0 \cdot 1) \cdot 138 \\ &= 222314.88 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (1511 + 0 + 100 + (629)) \cdot 138 \\ &= 309168.94 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (1511 + 0 + 100 + 0 + (629)) \cdot 138 \\ &= 309168.94 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$



$$= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138 \\ = 124631. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ = (3.14 \cdot 49.1) \cdot (20.6 - 1.5) \cdot 0.7 \cdot 138 \\ = 284402. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ = (3.14/2.0) \cdot 117 \cdot (18 - 1.5) \cdot 0.74 \cdot 138 \\ = 310560. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (124631 + 284402) = 409033 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw) \\ = (124631 + 0 + 310560 + 0) = 435191 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw) \\ = (124631 + 310560 + 0) = 435191 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 409032 N	, must exceed W = 176664 N	or W1 = 222314 N
Path 2-2 = 435191 N	, must exceed W = 176664 N	or W2 = 309168 N
Path 3-3 = 435191 N	, must exceed W = 176664 N	or W3 = 309168 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 46.7746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.9370 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K3C3"

From: 20

Pressure for Reinforcement Calculations	P	2.6024	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	500.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		2300.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		54.83	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K3C3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.000	in.
Actual Thickness Used in Calculation		0.810	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 16.2415 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.6 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 0.7553 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	194.9944	mm
Parallel to Vessel Wall, opening length	d	97.4972	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	791.750	521.230	NA
Area in Shell	A1	816.954	566.243	NA
Area in Nozzle Wall	A2	1782.732	1810.427	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	2699.686	2476.670	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 54.32 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (97.5 \cdot 16.2 \cdot 0.5 + 2 \cdot 19.1 \cdot 16.2 \cdot 0.5 \cdot (1 - 1))$$

$$= 791.750 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 97.5 (1 \cdot 16.5 - 0.5 \cdot 16.2) - 2 \cdot 19.1$$

$$(1 \cdot 16.5 - 0.5 \cdot 16.2) \cdot (1 - 1)$$

$$= 816.954 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 41.2) (19.1 - 0.76) / \sin(58)$$

$$= 1782.732 \text{ mm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.2, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K3C3"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (792 - 817 + 2 • 19.1 • 1 •
(1 • 16.5 - 8.12))138)
= 40629.63 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1783 + 0 + 100 - 0 • 1) • 138
= 259790.25 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1783 + 0 + 100 + (629)) • 138
= 346644.31 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1783 + 0 + 100 + 0 + (629)) • 138
= 346644.31 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 144 • 10 • 0.49 • 138
= 153424. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 60.5) • (20.6 - 1.5) • 0.7 • 138
= 350106. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 144 • (18 - 1.5) • 0.74 • 138
= 382307. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (153424 + 350106) = 503530 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (153424 + 0 + 382307 + 0) = 535731 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (153424 + 382307 + 0) = 535731 N

Summary of Failure Path Calculations:



Path 1-1 = 503529 N , must exceed W = 40629 N or W1 = 259790 N
 Path 2-2 = 535731 N , must exceed W = 40629 N or W2 = 346644 N
 Path 3-3 = 535731 N , must exceed W = 40629 N or W3 = 346644 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3C3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$
 $= 16.2415 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.6 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.6)$
 $= 0.7553 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 158.4000 mm
 Parallel to Vessel Wall, opening length d 79.2000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 41.2500 mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1286.326	423.412	NA
Area in Shell	A1	20.474	459.977	NA
Area in Nozzle Wall	A2	1511.290	1534.768	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1631.764	2094.745	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (79.2 \cdot 16.2 \cdot 1 + 2 \cdot 19.1 \cdot 16.2 \cdot 1 \cdot (1 - 1))$
 $= 1286.326 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2 (1 \cdot 16.5 - 1 \cdot 16.2) - 2 \cdot 19.1$
 $(1 \cdot 16.5 - 1 \cdot 16.2) \cdot (1 - 1)$



$$= 20.474 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot t_{np})(t_n - t_{rn})fr_2 \\ &= (2 \cdot 41.2)(19.1 - 0.76)1 \\ &= 1511.290 \text{ mm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.2553 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 17.7415 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 17.7415 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 2.1385 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb_3, \max(tb_1, tb_2)] \\ &= \min[7.23, \max(17.7, 3)] \\ &= 7.2300 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(2.26, 7.23) \\ &= 7.2300 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Weld Size Calculations, Description: K3C3"

Intermediate Calc. for nozzle/shell welds $T_{min} = 16.5000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v) \\ &= \max(0, (1286 - 20.5 + 2 \cdot 19.1 \cdot 1 \cdot (1 \cdot 16.5 - 16.2)) \cdot 138) \\ &= 176030.36 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_5 + A_4 - (W_i - Can/.707)^2 \cdot fr_2) \cdot S_v \\ &= (1511 + 0 + 100 - 0 \cdot 1) \cdot 138 \\ &= 222335.12 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (1511 + 0 + 100 + (629)) \cdot 138 \\ &= 309189.16 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (1511 + 0 + 100 + 0 + (629)) \cdot 138 \\ &= 309189.16 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$



$$= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138 \\ = 124631. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ = (3.14 \cdot 49.1) \cdot (20.6 - 1.5) \cdot 0.7 \cdot 138 \\ = 284402. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ = (3.14/2.0) \cdot 117 \cdot (18 - 1.5) \cdot 0.74 \cdot 138 \\ = 310560. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (124631 + 284402) = 409033 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw) \\ = (124631 + 0 + 310560 + 0) = 435191 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw) \\ = (124631 + 310560 + 0) = 435191 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 409032 N	, must exceed W = 176030 N	or W1 = 222335 N
Path 2-2 = 435191 N	, must exceed W = 176030 N	or W2 = 309189 N
Path 3-3 = 435191 N	, must exceed W = 176030 N	or W3 = 309189 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 46.7746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.9370 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K3D3"

From: 20

Pressure for Reinforcement Calculations	P	2.6084	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	500.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		2300.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		125.17	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K3D3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.000	in.
Actual Thickness Used in Calculation		0.810	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 16.2797 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.61 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 0.7571 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	194.9920	mm
Parallel to Vessel Wall, opening length	d	97.4960	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	793.604	521.224	NA
Area in Shell	A1	815.080	566.236	NA
Area in Nozzle Wall	A2	1782.559	1810.427	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	2697.639	2476.663	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 54.33 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (97.5 \cdot 16.3 \cdot 0.5 + 2 \cdot 19.1 \cdot 16.3 \cdot 0.5 \cdot (1 - 1))$$

$$= 793.604 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 97.5 (1 \cdot 16.5 - 0.5 \cdot 16.3) - 2 \cdot 19.1$$

$$(1 \cdot 16.5 - 0.5 \cdot 16.3) \cdot (1 - 1)$$

$$= 815.080 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 41.2) (19.1 - 0.76) / \sin(58)$$

$$= 1782.559 \text{ mm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.3, c = 1.5 mm, E* = 1

Thickness Ratio = tr * E*/(tg - c) = 0.99, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K3D3"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (794 - 815 + 2 • 19.1 • 1 •
(1 • 16.5 - 8.14))138)
= 41043.37 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1783 + 0 + 100 - 0 • 1) • 138
= 259766.34 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1783 + 0 + 100 + (629)) • 138
= 346620.41 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1783 + 0 + 100 + 0 + (629)) • 138
= 346620.41 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 144 • 10 • 0.49 • 138
= 153422. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 60.5) • (20.6 - 1.5) • 0.7 • 138
= 350102. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 144 • (18 - 1.5) • 0.74 • 138
= 382302. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (153422 + 350102) = 503524 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (153422 + 0 + 382302 + 0) = 535724 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (153422 + 382302 + 0) = 535724 N

Summary of Failure Path Calculations:



Path 1-1 = 503523 N , must exceed W = 41043 N or W1 = 259766 N
 Path 2-2 = 535724 N , must exceed W = 41043 N or W2 = 346620 N
 Path 3-3 = 535724 N , must exceed W = 41043 N or W3 = 346620 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3D3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$
 $= 16.2797 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.61 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.61)$
 $= 0.7571 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 158.4000 mm
 Parallel to Vessel Wall, opening length d 79.2000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 41.2500 mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1289.354	423.412	NA
Area in Shell	A1	17.446	459.977	NA
Area in Nozzle Wall	A2	1511.144	1534.768	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1628.589	2094.745	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (79.2 \cdot 16.3 \cdot 1 + 2 \cdot 19.1 \cdot 16.3 \cdot 1 \cdot (1 - 1))$
 $= 1289.354 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2 (1 \cdot 16.5 - 1 \cdot 16.3) - 2 \cdot 19.1$
 $(1 \cdot 16.5 - 1 \cdot 16.3) \cdot (1 - 1)$



$$= 17.446 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot t_{np})(t_n - t_{rn})fr_2 \\ &= (2 \cdot 41.2)(19.1 - 0.76)1 \\ &= 1511.144 \text{ mm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.2571 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 17.7797 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 17.7797 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 2.1385 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb_3, \max(tb_1, tb_2)] \\ &= \min[7.23, \max(17.8, 3)] \\ &= 7.2300 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(2.26, 7.23) \\ &= 7.2300 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Weld Size Calculations, Description: K3D3"

Intermediate Calc. for nozzle/shell welds $T_{min} = 16.5000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v) \\ &= \max(0, (1289 - 17.4 + 2 \cdot 19.1 \cdot 1 \cdot (1 \cdot 16.5 - 16.3)) \cdot 138) \\ &= 176664.73 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_5 + A_4 - (W_i - Can/.707)^2 \cdot fr_2) \cdot S_v \\ &= (1511 + 0 + 100 - 0 \cdot 1) \cdot 138 \\ &= 222314.88 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (1511 + 0 + 100 + (629)) \cdot 138 \\ &= 309168.94 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (1511 + 0 + 100 + 0 + (629)) \cdot 138 \\ &= 309168.94 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$



$$= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138 \\ = 124631. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ = (3.14 \cdot 49.1) \cdot (20.6 - 1.5) \cdot 0.7 \cdot 138 \\ = 284402. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ = (3.14/2.0) \cdot 117 \cdot (18 - 1.5) \cdot 0.74 \cdot 138 \\ = 310560. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (124631 + 284402) = 409033 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw) \\ = (124631 + 0 + 310560 + 0) = 435191 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw) \\ = (124631 + 310560 + 0) = 435191 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 409032 N	, must exceed W = 176664 N	or W1 = 222314 N
Path 2-2 = 435191 N	, must exceed W = 176664 N	or W2 = 309168 N
Path 3-3 = 435191 N	, must exceed W = 176664 N	or W3 = 309168 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 46.7746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.9370 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K3E3"

From: 20

Pressure for Reinforcement Calculations	P	2.6024	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	500.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		2600.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		54.83	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K3E3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.000	in.
Actual Thickness Used in Calculation		0.810	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 16.2415 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.6 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 0.7553 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	194.9944	mm
Parallel to Vessel Wall, opening length	d	97.4972	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	791.750	521.230	NA
Area in Shell	A1	816.954	566.243	NA
Area in Nozzle Wall	A2	1782.732	1810.427	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	2699.686	2476.670	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 54.32 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (97.5 \cdot 16.2 \cdot 0.5 + 2 \cdot 19.1 \cdot 16.2 \cdot 0.5 \cdot (1 - 1))$$

$$= 791.750 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 97.5 (1 \cdot 16.5 - 0.5 \cdot 16.2) - 2 \cdot 19.1$$

$$(1 \cdot 16.5 - 0.5 \cdot 16.2) \cdot (1 - 1)$$

$$= 816.954 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 41.2) (19.1 - 0.76) / \sin(58)$$

$$= 1782.732 \text{ mm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.2, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K3E3"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (792 - 817 + 2 • 19.1 • 1 •
(1 • 16.5 - 8.12))138)
= 40629.63 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1783 + 0 + 100 - 0 • 1) • 138
= 259790.25 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1783 + 0 + 100 + (629)) • 138
= 346644.31 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1783 + 0 + 100 + 0 + (629)) • 138
= 346644.31 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 144 • 10 • 0.49 • 138
= 153424. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 60.5) • (20.6 - 1.5) • 0.7 • 138
= 350106. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 144 • (18 - 1.5) • 0.74 • 138
= 382307. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (153424 + 350106) = 503530 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (153424 + 0 + 382307 + 0) = 535731 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (153424 + 382307 + 0) = 535731 N

Summary of Failure Path Calculations:



Path 1-1 = 503529 N , must exceed W = 40629 N or W1 = 259790 N
 Path 2-2 = 535731 N , must exceed W = 40629 N or W2 = 346644 N
 Path 3-3 = 535731 N , must exceed W = 40629 N or W3 = 346644 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3E3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$
 $= 16.2415 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.6 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.6)$
 $= 0.7553 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 158.4000 mm
 Parallel to Vessel Wall, opening length d 79.2000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 41.2500 mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1286.326	423.412	NA
Area in Shell	A1	20.474	459.977	NA
Area in Nozzle Wall	A2	1511.290	1534.768	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1631.764	2094.745	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (79.2 \cdot 16.2 \cdot 1 + 2 \cdot 19.1 \cdot 16.2 \cdot 1 \cdot (1 - 1))$
 $= 1286.326 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2 (1 \cdot 16.5 - 1 \cdot 16.2) - 2 \cdot 19.1$
 $(1 \cdot 16.5 - 1 \cdot 16.2) \cdot (1 - 1)$



$$= 20.474 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot t_{np})(t_n - t_r)fr_2 \\ &= (2 \cdot 41.2)(19.1 - 0.76)1 \\ &= 1511.290 \text{ mm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2553 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.7415 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.7415 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.1385 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 7.2300 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[7.23, \max(17.7, 3)] \\ &= 7.2300 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(2.26, 7.23) \\ &= 7.2300 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Weld Size Calculations, Description: K3E3"

Intermediate Calc. for nozzle/shell welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr))Sv) \\ &= \max(0, (1286 - 20.5 + 2 \cdot 19.1 \cdot 1 \cdot (1 \cdot 16.5 - 16.2))138) \\ &= 176030.36 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A5 + A4 - (W_i - Can/.707)^2 \cdot fr_2) \cdot Sv \\ &= (1511 + 0 + 100 - 0 \cdot 1) \cdot 138 \\ &= 222335.12 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv \\ &= (1511 + 0 + 100 + (629)) \cdot 138 \\ &= 309189.16 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (1511 + 0 + 100 + 0 + (629)) \cdot 138 \\ &= 309189.16 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$$



$$= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138 \\ = 124631. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ = (3.14 \cdot 49.1) \cdot (20.6 - 1.5) \cdot 0.7 \cdot 138 \\ = 284402. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ = (3.14/2.0) \cdot 117 \cdot (18 - 1.5) \cdot 0.74 \cdot 138 \\ = 310560. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (124631 + 284402) = 409033 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw) \\ = (124631 + 0 + 310560 + 0) = 435191 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw) \\ = (124631 + 310560 + 0) = 435191 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 409032 N	, must exceed W = 176030 N	or W1 = 222335 N
Path 2-2 = 435191 N	, must exceed W = 176030 N	or W2 = 309189 N
Path 3-3 = 435191 N	, must exceed W = 176030 N	or W3 = 309189 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 46.7746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.9370 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K3F3"

From: 20

Pressure for Reinforcement Calculations	P	2.6084	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	500.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		2600.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		125.17	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K3F3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.000	in.
Actual Thickness Used in Calculation		0.810	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 16.2797 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.61 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 0.7571 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	194.9920	mm
Parallel to Vessel Wall, opening length	d	97.4960	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	793.604	521.224	NA
Area in Shell	A1	815.080	566.236	NA
Area in Nozzle Wall	A2	1782.559	1810.427	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	2697.639	2476.663	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 54.33 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (97.5 \cdot 16.3 \cdot 0.5 + 2 \cdot 19.1 \cdot 16.3 \cdot 0.5 \cdot (1 - 1))$$

$$= 793.604 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 97.5 (1 \cdot 16.5 - 0.5 \cdot 16.3) - 2 \cdot 19.1$$

$$(1 \cdot 16.5 - 0.5 \cdot 16.3) \cdot (1 - 1)$$

$$= 815.080 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 41.2) (19.1 - 0.76) / \sin(58)$$

$$= 1782.559 \text{ mm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.3, c = 1.5 mm, E* = 1

Thickness Ratio = tr * E*/(tg - c) = 0.99, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K3F3"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (794 - 815 + 2 • 19.1 • 1 •
(1 • 16.5 - 8.14))138)
= 41043.37 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1783 + 0 + 100 - 0 • 1) • 138
= 259766.34 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1783 + 0 + 100 + (629)) • 138
= 346620.41 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1783 + 0 + 100 + 0 + (629)) • 138
= 346620.41 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 144 • 10 • 0.49 • 138
= 153422. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 60.5) • (20.6 - 1.5) • 0.7 • 138
= 350102. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 144 • (18 - 1.5) • 0.74 • 138
= 382302. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (153422 + 350102) = 503524 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (153422 + 0 + 382302 + 0) = 535724 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (153422 + 382302 + 0) = 535724 N

Summary of Failure Path Calculations:



Path 1-1 = 503523 N , must exceed W = 41043 N or W1 = 259766 N
 Path 2-2 = 535724 N , must exceed W = 41043 N or W2 = 346620 N
 Path 3-3 = 535724 N , must exceed W = 41043 N or W3 = 346620 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3F3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$
 $= 16.2797 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.61 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.61)$
 $= 0.7571 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 158.4000 mm
 Parallel to Vessel Wall, opening length d 79.2000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 41.2500 mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1289.354	423.412	NA
Area in Shell	A1	17.446	459.977	NA
Area in Nozzle Wall	A2	1511.144	1534.768	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1628.589	2094.745	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (79.2 \cdot 16.3 \cdot 1 + 2 \cdot 19.1 \cdot 16.3 \cdot 1 \cdot (1 - 1))$
 $= 1289.354 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2 (1 \cdot 16.5 - 1 \cdot 16.3) - 2 \cdot 19.1$
 $(1 \cdot 16.5 - 1 \cdot 16.3) \cdot (1 - 1)$



$$= 17.446 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot t_{np})(t_n - t_{rn})fr_2 \\ &= (2 \cdot 41.2)(19.1 - 0.76)1 \\ &= 1511.144 \text{ mm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2571 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.7797 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.7797 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.1385 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 7.2300 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[7.23, \max(17.8, 3)] \\ &= 7.2300 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(2.26, 7.23) \\ &= 7.2300 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Weld Size Calculations, Description: K3F3"

Intermediate Calc. for nozzle/shell welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv) \\ &= \max(0, (1289 - 17.4 + 2 \cdot 19.1 \cdot 1 \cdot (1 \cdot 16.5 - 16.3)) \cdot 138) \\ &= 176664.73 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2 + A5 + A4 - (W_i - Can/.707)^2 \cdot fr_2) \cdot Sv \\ &= (1511 + 0 + 100 - 0 \cdot 1) \cdot 138 \\ &= 222314.88 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv \\ &= (1511 + 0 + 100 + (629)) \cdot 138 \\ &= 309168.94 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (1511 + 0 + 100 + 0 + (629)) \cdot 138 \\ &= 309168.94 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$$



$$= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138 \\ = 124631. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ = (3.14 \cdot 49.1) \cdot (20.6 - 1.5) \cdot 0.7 \cdot 138 \\ = 284402. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ = (3.14/2.0) \cdot 117 \cdot (18 - 1.5) \cdot 0.74 \cdot 138 \\ = 310560. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (124631 + 284402) = 409033 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw) \\ = (124631 + 0 + 310560 + 0) = 435191 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw) \\ = (124631 + 310560 + 0) = 435191 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 409032 N	, must exceed W = 176664 N	or W1 = 222314 N
Path 2-2 = 435191 N	, must exceed W = 176664 N	or W2 = 309168 N
Path 3-3 = 435191 N	, must exceed W = 176664 N	or W3 = 309168 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 46.7746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.9370 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K61"

From: 20

Pressure for Reinforcement Calculations	P	2.6002	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		580.00	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	426.2520	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K61"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.000	in.
Actual Thickness Used in Calculation	0.560	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$
 $= 16.2274 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 14.2 \cdot (\exp([2.6 / (138 \cdot 1)] - 1))$
 $= 0.2701 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4180 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	86.8480 mm
Parallel to Vessel Wall	Rn+tn+tt	43.4240 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	31.8100 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K61".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9180 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.7274 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.7274 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.1385 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 4.9200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(17.7, 3)]$
 $= 4.9200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.92, 4.92)$
 $= 4.9200 \text{ mm}$

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 14.2, tr = 0.27, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.021$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K61"

Intermediate Calc. for nozzle/shell Welds	Tmin	12.7240 mm
---	------	------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*



The Drop for this Nozzle is : 0.4265 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 444.6785 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 43.42 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N52"

From: 20

Pressure for Reinforcement Calculations	P	2.6107	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		500.00	mm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	236.6000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N52"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$
 = 16.2938 mm



Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R(\exp([P/(Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $26.9(\exp([2.61/(138 \cdot 1)] - 1))$
= 0.5137 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4168 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	117.3280	mm
Parallel to Vessel Wall	Rn+tn+t	58.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N52".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0137	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.7938	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.7938	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.1385	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[6.3, max(17.8, 3)]
= 6.3000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(2.01, 6.3)
= 6.3000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
(see 319.4.4 and 302.3.5):**

Sustained	:	11.6,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	333.4 MPa	Passed
Occasional	:	1.8,	Allowable	:	183.5 MPa	Passed
Shear	:	9.4,	Allowable	:	96.6 MPa	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
the expansion stress allowable.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.51, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.034$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N52"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640	mm
---	------	---------	----

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0464 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 255.6463 mm

Input Echo, WRC297 Item 1, Description: N52" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.5000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1700.000	mm	
Shell Thickness	18.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	1.5000	mm	
Nozzle Diameter	50.800	mm	
Nozzle Thickness	16.7640	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-350 LF2		
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	2.61	MPa
Radial Load	P	-1300.00	N
Circumferential Shear	Vc	2000.00	N
Longitudinal Shear	Vl	2000.00	N
Circumferential Moment	Mc	300000.00	N-mm
Longitudinal Moment	Ml	400000.03	N-mm
Torsional Moment	Mt	500000.00	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (4.685) is < 5
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (5.030) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(859.75(18.0 - 1.5))}$$

$$= 477.373 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1719.500 mm
Nozzle Outside Diameter	(d) =	84.328 mm
Vessel Thickness used	(T) =	16.500 mm
Nozzle Thickness used	(t) =	15.264 mm
T / t	=	1.081



$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.501$$

Nr/P	=	0.178
Mr/P	=	0.272
M0/P	=	0.100
N0/P	=	0.172
MrD/Mc	=	0.285
NrDL/Mc	=	0.148
M0d/Mc	=	0.122
N0DL/Mc	=	-0.101
MrD/Ml	=	0.257
NrDL/Ml	=	0.143
M0D/Ml	=	0.107
N0DL/Ml	=	-0.015

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-11	11	11	-11
Normal Pressure Stress	133	136	133	136
Outplane Stress Summary	126	145	148	122

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-26	26	26	-26
Inplane Pressure Stress	66	66	66	66
Inplane Stress Summary	46	84	104	35

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	126	145	148	122

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-9	9	9	-9
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	66	66	66	66

Outplane Stress Summary	62	75	78	53
-------------------------	----	----	----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-22	22	22	-22
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	133	136	133	136

Inplane Stress Summary	117	149	166	108
------------------------	-----	-----	-----	-----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	117	149	166	108
------------------------	-----	-----	-----	-----

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7

Hoop Stress Summary	5	8	4	7
---------------------	---	---	---	---

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-22	22	22	-22
Axial Pressure Stress	2	2	2	2



Axial Stress Summary -22 10 39 -18

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary 3 3 1 1

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	28	13	39	26

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7

Hoop Stress Summary 6 9 3 6

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-19	19	19	-19
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary -16 9 34 -16

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary 1 1 3 3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	23	11	34	24

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of	Stress Values at
---------	------------------

Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	133	136	133	136	133	136	133	136
Circ. Pl (SUS)	0	0	0	0	-1	-1	1	1
Circ. Q (SUS)	-9	9	13	-13	-15	15	29	-29
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	-2	-2	2	2	1	1	-1	-1
Long. Q (SUS)	-19	19	33	-33	-7	7	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	133.0	136.0	133.0	136.0	133.0	136.0	133.0	136.0
Pm+Pl (SUS)	133.0	136.0	133.0	136.0	132.0	135.0	134.0	137.0
Pm+Pl+Q (Total)	124.1	145.1	146.1	123.0	117.1	150.1	163.0	108.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+Pl (SUS)	137.00	207.00	Passed
Pm+Pl+Q (TOTAL)	163.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	7	4	7	4	7	4	7
Circ. Pl (SUS)	0	0	0	0	1	1	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-8	-8	8	8	-6	-6	6	6
Long. Q (SUS)	-16	16	28	-28	-13	13	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	4.0	7.0	4.0	7.0	4.0	7.0	4.0	7.0
Pm+Pl (SUS)	10.0	13.0	10.0	10.0	9.0	12.0	8.0	8.0
Pm+Pl+Q (Total)	26.3	11.0	38.1	25.3	22.4	10.6	33.1	23.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	138.00	Passed
Pm+Pl (SUS)	13.00	207.00	Passed
Pm+Pl+Q (TOTAL)	38.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 Licensee: SL User
FileName : V-31002_7%
Nozzle Calcs.: N5_2"

Page 161 of 218
May 20, 2026



HEXAGON

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K1A3"

From: 30

Pressure for Reinforcement Calculations	P	2.6024	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	500.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		4700.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		54.83	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K1A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.000	in.
Actual Thickness Used in Calculation		0.810	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 16.2415 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.6 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 0.7553 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	194.9944	mm
Parallel to Vessel Wall, opening length	d	97.4972	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	791.750	521.230	NA
Area in Shell	A1	816.954	566.243	NA
Area in Nozzle Wall	A2	1782.732	1810.427	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	2699.686	2476.670	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 54.32 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (97.5 \cdot 16.2 \cdot 0.5 + 2 \cdot 19.1 \cdot 16.2 \cdot 0.5 \cdot (1 - 1))$$

$$= 791.750 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 97.5 (1 \cdot 16.5 - 0.5 \cdot 16.2) - 2 \cdot 19.1$$

$$(1 \cdot 16.5 - 0.5 \cdot 16.2) \cdot (1 - 1)$$

$$= 816.954 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 41.2) (19.1 - 0.76) / \sin(58)$$

$$= 1782.732 \text{ mm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.2, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E* / (tg - c) = 0.98, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K1A3"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (792 - 817 + 2 • 19.1 • 1 •
(1 • 16.5 - 8.12))138)
= 40629.63 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1783 + 0 + 100 - 0 • 1) • 138
= 259790.25 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1783 + 0 + 100 + (629)) • 138
= 346644.31 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1783 + 0 + 100 + 0 + (629)) • 138
= 346644.31 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 144 • 10 • 0.49 • 138
= 153424. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 60.5) • (20.6 - 1.5) • 0.7 • 138
= 350106. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 144 • (18 - 1.5) • 0.74 • 138
= 382307. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (153424 + 350106) = 503530 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (153424 + 0 + 382307 + 0) = 535731 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (153424 + 382307 + 0) = 535731 N

Summary of Failure Path Calculations:



Path 1-1 = 503529 N , must exceed W = 40629 N or W1 = 259790 N
 Path 2-2 = 535731 N , must exceed W = 40629 N or W2 = 346644 N
 Path 3-3 = 535731 N , must exceed W = 40629 N or W3 = 346644 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K1A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$
 $= 16.2415 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.6 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.6)$
 $= 0.7553 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 158.4000 mm
 Parallel to Vessel Wall, opening length d 79.2000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 41.2500 mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1286.326	423.412	NA
Area in Shell	A1	20.474	459.977	NA
Area in Nozzle Wall	A2	1511.290	1534.768	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1631.764	2094.745	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (79.2 \cdot 16.2 \cdot 1 + 2 \cdot 19.1 \cdot 16.2 \cdot 1 \cdot (1 - 1))$
 $= 1286.326 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2 (1 \cdot 16.5 - 1 \cdot 16.2) - 2 \cdot 19.1$
 $(1 \cdot 16.5 - 1 \cdot 16.2) \cdot (1 - 1)$



$$= 20.474 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot t_{np})(t_n - t_{rn})fr_2 \\ &= (2 \cdot 41.2)(19.1 - 0.76)1 \\ &= 1511.290 \text{ mm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.2553 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 17.7415 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 17.7415 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 2.1385 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb_3, \max(tb_1, tb_2)] \\ &= \min[7.23, \max(17.7, 3)] \\ &= 7.2300 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(2.26, 7.23) \\ &= 7.2300 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Weld Size Calculations, Description: K1A3"

Intermediate Calc. for nozzle/shell welds	T_{min}	16.5000 mm
---	-----------	----------------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v) \\ &= \max(0, (1286 - 20.5 + 2 \cdot 19.1 \cdot 1 \cdot (1 \cdot 16.5 - 16.2)) \cdot 138) \\ &= 176030.36 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_5 + A_4 - (W_i - Can/.707)^2 \cdot fr_2) \cdot S_v \\ &= (1511 + 0 + 100 - 0 \cdot 1) \cdot 138 \\ &= 222335.12 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (1511 + 0 + 100 + (629)) \cdot 138 \\ &= 309189.16 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (1511 + 0 + 100 + 0 + (629)) \cdot 138 \\ &= 309189.16 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$



$$= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138 \\ = 124631. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ = (3.14 \cdot 49.1) \cdot (20.6 - 1.5) \cdot 0.7 \cdot 138 \\ = 284402. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ = (3.14/2.0) \cdot 117 \cdot (18 - 1.5) \cdot 0.74 \cdot 138 \\ = 310560. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (124631 + 284402) = 409033 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw) \\ = (124631 + 0 + 310560 + 0) = 435191 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw) \\ = (124631 + 310560 + 0) = 435191 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 409032 N	, must exceed W = 176030 N	or W1 = 222335 N
Path 2-2 = 435191 N	, must exceed W = 176030 N	or W2 = 309189 N
Path 3-3 = 435191 N	, must exceed W = 176030 N	or W3 = 309189 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 46.7746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.9370 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K1B3"

From: 30

Pressure for Reinforcement Calculations	P	2.6084	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	500.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		4700.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		125.17	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K1B3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		3.000	in.
Actual Thickness Used in Calculation		0.810	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 16.2797 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.61 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 0.7571 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	194.9920	mm
Parallel to Vessel Wall, opening length	d	97.4960	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	793.604	521.224	NA
Area in Shell	A1	815.080	566.236	NA
Area in Nozzle Wall	A2	1782.559	1810.427	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	2697.639	2476.663	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 54.33 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (97.5 \cdot 16.3 \cdot 0.5 + 2 \cdot 19.1 \cdot 16.3 \cdot 0.5 \cdot (1 - 1))$$

$$= 793.604 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 97.5 (1 \cdot 16.5 - 0.5 \cdot 16.3) - 2 \cdot 19.1$$

$$(1 \cdot 16.5 - 0.5 \cdot 16.3) \cdot (1 - 1)$$

$$= 815.080 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$$

$$= (2 \cdot 41.2) (19.1 - 0.76) / \sin(58)$$

$$= 1782.559 \text{ mm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.3, c = 1.5 mm, E* = 1

Thickness Ratio = tr * E*/(tg - c) = 0.99, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K1B3"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (794 - 815 + 2 • 19.1 • 1 •
(1 • 16.5 - 8.14))138)
= 41043.37 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1783 + 0 + 100 - 0 • 1) • 138
= 259766.34 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1783 + 0 + 100 + (629)) • 138
= 346620.41 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1783 + 0 + 100 + 0 + (629)) • 138
= 346620.41 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 144 • 10 • 0.49 • 138
= 153422. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 60.5) • (20.6 - 1.5) • 0.7 • 138
= 350102. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 144 • (18 - 1.5) • 0.74 • 138
= 382302. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (153422 + 350102) = 503524 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (153422 + 0 + 382302 + 0) = 535724 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (153422 + 382302 + 0) = 535724 N

Summary of Failure Path Calculations:



Path 1-1 = 503523 N , must exceed W = 41043 N or W1 = 259766 N
 Path 2-2 = 535724 N , must exceed W = 41043 N or W2 = 346620 N
 Path 3-3 = 535724 N , must exceed W = 41043 N or W3 = 346620 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K1B3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$
 $= 16.2797$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.61 \cdot 39.6 / (138 \cdot 1 - 0.6 \cdot 2.61)$
 $= 0.7571$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4707 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 158.4000 mm
 Parallel to Vessel Wall, opening length d 79.2000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 41.2500 mm

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1289.354	423.412	NA
Area in Shell	A1	17.446	459.977	NA
Area in Nozzle Wall	A2	1511.144	1534.768	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1628.589	2094.745	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (79.2 \cdot 16.3 \cdot 1 + 2 \cdot 19.1 \cdot 16.3 \cdot 1 \cdot (1 - 1))$
 $= 1289.354$ mm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 79.2 (1 \cdot 16.5 - 1 \cdot 16.3) - 2 \cdot 19.1$
 $(1 \cdot 16.5 - 1 \cdot 16.3) \cdot (1 - 1)$



$$= 17.446 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot t_{np})(t_n - t_{rn})fr_2 \\ &= (2 \cdot 41.2)(19.1 - 0.76)1 \\ &= 1511.144 \text{ mm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.2571 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 17.7797 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 17.7797 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 2.1385 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 7.2300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb_3, \max(tb_1, tb_2)] \\ &= \min[7.23, \max(17.8, 3)] \\ &= 7.2300 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(2.26, 7.23) \\ &= 7.2300 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Weld Size Calculations, Description: K1B3"

Intermediate Calc. for nozzle/shell welds $T_{min} = 16.5000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v) \\ &= \max(0, (1289 - 17.4 + 2 \cdot 19.1 \cdot 1 \cdot (1 \cdot 16.5 - 16.3)) \cdot 138) \\ &= 176664.73 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_5 + A_4 - (W_i - Can/.707)^2 \cdot fr_2) \cdot S_v \\ &= (1511 + 0 + 100 - 0 \cdot 1) \cdot 138 \\ &= 222314.88 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v \\ &= (1511 + 0 + 100 + (629)) \cdot 138 \\ &= 309168.94 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S \\ &= (1511 + 0 + 100 + 0 + (629)) \cdot 138 \\ &= 309168.94 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$$



$$= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138$$
$$= 124631. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.14 \cdot 49.1) \cdot (20.6 - 1.5) \cdot 0.7 \cdot 138$$
$$= 284402. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$
$$= (3.14/2.0) \cdot 117 \cdot (18 - 1.5) \cdot 0.74 \cdot 138$$
$$= 310560. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (124631 + 284402) = 409033 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (124631 + 0 + 310560 + 0) = 435191 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$
$$= (124631 + 310560 + 0) = 435191 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 409032 N	, must exceed W = 176664 N	or W1 = 222314 N
Path 2-2 = 435191 N	, must exceed W = 176664 N	or W2 = 309168 N
Path 3-3 = 435191 N	, must exceed W = 176664 N	or W3 = 309168 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 46.7746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.9370 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K2A1.5"

From: 30

Pressure for Reinforcement Calculations	P	2.6024	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	500.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		4500.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		54.83	deg
Diameter		1.5000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	207.9620	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K2A1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		1.500	in.
Actual Thickness Used in Calculation		0.630	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the



attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$
= 16.2415 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $20.5 \cdot (\exp([2.6 / (138 \cdot 1)] - 1))$
= 0.3912 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3545 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	112.5488	mm
Parallel to Vessel Wall	Rn+tn+t	56.2744	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2A1.5".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16, tr = 0.39, c = 1.5 mm, $E^* = 1$
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K2A1.5"

Intermediate Calc. for nozzle/shell Welds Tmin 14.5020 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K2A1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500 in.
Actual Thickness Used in Calculation	0.630 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 16.2415 \text{ mm}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R (\exp [P / (S_n \cdot E)] - 1) \text{ per Appendix 1-2 (a)(1)}$$

$$= 20.5 (\exp [2.6 / (138 \cdot 1)] - 1)$$

$$= 0.3912 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3545 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	103.1040	mm
Parallel to Vessel Wall	Rn+tn+t	51.5520	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K2A1.5".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8912	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.7415	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.7415	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.1385	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.0200	mm

Determine Nozzle Thickness candidate [tb]:

$$= \min [tb3, \max (tb1, tb2)]$$

$$= \min [6.02, \max (17.7, 3)]$$

$$= 6.0200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max (ta, tb)$$

$$= \max (1.89, 6.02)$$

$$= 6.0200 \text{ mm}$$

Available Nozzle Neck Thickness = 16.0020 mm -> OK

Weld Size Calculations, Description: K2A1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	14.5020	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 26.9173 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 257.0167 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K2B1.5"

From: 30

Pressure for Reinforcement Calculations	P	2.6084	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	500.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		4500.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		125.17	deg
Diameter		1.5000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	207.9620	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K2B1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		1.500	in.
Actual Thickness Used in Calculation		0.630	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the



attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$
= 16.2797 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $20.5 \cdot (\exp([2.61 / (138 \cdot 1)] - 1))$
= 0.3921 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3545 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	112.5482	mm
Parallel to Vessel Wall	Rn+tn+t	56.2741	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2B1.5".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govern. thk, tg = 16, tr = 0.39, c = 1.5 mm, $E^* = 1$
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K2B1.5"

Intermediate Calc. for nozzle/shell Welds Tmin 14.5020 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K2B1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500 in.
Actual Thickness Used in Calculation	0.630 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$

= 16.2797 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

= $R (\exp [P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)

= $20.5 (\exp [2.61 / (138 \cdot 1)] - 1)$

= 0.3921 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3545 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 103.1040 mm

Parallel to Vessel Wall Rn+tn+t 51.5520 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 36.2550 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K2E1.5".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 1.8921 mm

Wall Thickness per UG16(b), tr16b = 3.0000 mm

Wall Thickness, shell/head, internal pressure trb1 = 17.7797 mm

Wall Thickness tb1 = max(trb1, tr16b) = 17.7797 mm

Wall Thickness, shell/head, external pressure trb2 = 2.1385 mm

Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm

Wall Thickness per table UG-45 tb3 = 6.0200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]

= min[6.02, max(17.8, 3)]

= 6.0200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(1.89, 6.02)

= 6.0200 mm

Available Nozzle Neck Thickness = 16.0020 mm -> OK

Weld Size Calculations, Description: K2B1.5"

Intermediate Calc. for nozzle/shell Welds Tmin 14.5020 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 26.9173 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 257.0167 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K51.5"

From: 30

Pressure for Reinforcement Calculations	P	2.6066	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	200.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		5100.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		103.32	deg
Diameter		1.5000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	207.9620	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K51.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		1.500	in.
Actual Thickness Used in Calculation		0.630	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the



attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$
= 16.2683 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $20.5 \cdot (\exp([2.61 / (138 \cdot 1)] - 1))$
= 0.3919 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3545 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	104.2686	mm
Parallel to Vessel Wall	Rn+tn+t	52.1343	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K51.5".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govern. thk, tg = 16, tr = 0.39, c = 1.5 mm, $E^* = 1$
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K51.5"

Intermediate Calc. for nozzle/shell Welds Tmin 14.5020 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K51.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500 in.
Actual Thickness Used in Calculation	0.630 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$
= 16.2683 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp \left[\frac{P}{S_n \cdot E} \right] - 1 \right)$ per Appendix 1-2 (a)(1)
= $20.5 \left(\exp \left[\frac{2.61}{138 \cdot 1} \right] - 1 \right)$
= 0.3919 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3545 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	103.1040	mm
Parallel to Vessel Wall	Rn+tn+t	51.5520	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K51.5"
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8919	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.7683	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.7683	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.1385	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.0200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.02, max(17.8, 3)]
= 6.0200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.89, 6.02)
= 6.0200 mm

Available Nozzle Neck Thickness = 16.0020 mm -> OK

Weld Size Calculations, Description: K51.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	14.5020	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 9.2815 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 235.7522 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K41.5"

From: 30

Pressure for Reinforcement Calculations	P	2.6066	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	200.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		4900.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		103.32	deg
Diameter		1.5000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0020	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	207.9620	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K41.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		1.500	in.
Actual Thickness Used in Calculation		0.630	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$\begin{aligned} &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61) \\ &= 16.2683 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Reqd thk per App. 1 of Nozzle Wall, trn} \quad [\text{Int. Press}] \\ &= R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right) \text{ per Appendix 1-2 (a)(1)} \\ &= 20.5 \left(\exp \left(\frac{2.61}{138 \cdot 1} \right) - 1 \right) \\ &= 0.3919 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3545 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	104.2686	mm
Parallel to Vessel Wall	Rn+tn+t	52.1343	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K41.5".
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16, tr = 0.39, c = 1.5 mm, E* = 1
Thickness Ratio = tr * E* / (tg - c) = 0.027, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: K41.5"

Intermediate Calc. for nozzle/shell Welds Tmin 14.5020 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K41.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500 in.
Actual Thickness Used in Calculation	0.630 in.

Nozzle input data check completed without errors.

$$\begin{aligned} &\text{Reqd thk per UG-37(a) of Cylindrical Shell, tr} \quad [\text{Int. Press}] \\ &= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61) \\ &= 16.2683 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Reqd thk per App. 1 of Nozzle Wall, trn} \quad [\text{Int. Press}] \\ &= R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right) \text{ per Appendix 1-2 (a)(1)} \\ &= 20.5 \left(\exp \left(\frac{2.61}{138 \cdot 1} \right) - 1 \right) \\ &= 0.3919 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.3545 mm



UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	103.1040	mm
Parallel to Vessel Wall	Rn+tn+t	51.5520	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.2550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K41.5".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8919	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.7683	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.7683	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.1385	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.0200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.02, max(17.8, 3)]
= 6.0200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.89, 6.02)
= 6.0200 mm

Available Nozzle Neck Thickness = 16.0020 mm -> OK

Weld Size Calculations, Description: K41.5"

Intermediate Calc. for nozzle/shell Welds Tmin 14.5020 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 9.2815 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 235.7522 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N36"

From: 40

Pressure for Reinforcement Calculations	P	2.6107	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7690.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-350 LF2 +IT	
-----------------	--	----------------	--

Hub Height of Integral Nozzle	h	40.0000	mm
Height of Beveled Transition	L'	43.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		54.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	164.9720	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		5.761	in.
Actual Thickness Used in Calculation		0.432	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)



$$= 2.61 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 16.2938 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.61 \cdot 74.7 / (138 \cdot 1 - 0.6 \cdot 2.61)$
 $= 1.4287 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5352 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 43 / 0.5773$
 $= 74.5253 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	298.6588	mm
Parallel to Vessel Wall, opening length	d	149.3294	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2433.143	798.331	NA
Area in Shell	A1	30.793	867.274	NA
Area in Nozzle Wall	A2	663.635	737.352	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	3548.183	3548.183	NA
TOTAL AREA AVAILABLE	Atot	4342.611	5252.809	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (149 \cdot 16.3 \cdot 1 + 2 \cdot 52.5 \cdot 16.3 \cdot 1 \cdot (1 - 1))$
 $= 2433.143 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:



$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 149(1 \cdot 16.5 - 1 \cdot 16.3) - 2 \cdot 52.5 \\ &\quad (1 \cdot 16.5 - 1 \cdot 16.3) \cdot (1 - 1) \\ &= 30.793 \text{ mm}^2 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot tlnp)(tn - trn)fr2 \\ &= (2 \cdot 41.2)(9.47 - 1.43)1 \\ &= 663.635 \text{ mm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\ &= (2 \cdot \min(41.2, 165, 40)) \cdot (54 - 11) \cdot 1 \\ &= 3548.183 \text{ mm}^2 \text{ Includes Bevel Area of } 106.007 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.9287 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.7938 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.7938 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.1385 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 7.7200 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[7.72, \max(17.8, 3)] \\ &= 7.7200 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(2.93, 7.72) \\ &= 7.7200 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 10.973 = 9.601 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 34.1,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 310.8 MPa	Passed
Occasional	: 9.7,	Allowable	: 183.5 MPa	Passed
Shear	: 17.4,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.3, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C

Flange MDMT with Temp reduction per UCS-66(i)(2) -76 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = 2.61/5.10 = 0.512

Weld Size Calculations, Description: N36"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (2433 - 30.8 + 2 • 52.5 • 1 •
(1 • 16.5 - 16.3))138)
= 334477.72 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (4212 + 100 - 0 • 1) • 138
= 594969.56 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (4212 + 0 + 100 + (1732)) • 138
= 834029.81 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (4212 + 0 + 100 + 0 + (1732)) • 138
= 834029.81 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 254 • 10 • 0.49 • 138
= 270114. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 101) • (54 - 1.5) • 0.7 • 138
= 1607666. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 254 • (18 - 1.5) • 0.74 • 138
= 673080. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (270114 + 1607666) = 1877780 N



Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (270114 + 0 + 673080 + 0) = 943193 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$
$$= (270114 + 673080 + 0) = 943193 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 1877780 N , must exceed W = 334477 N or W1 = 594969 N
Path 2-2 = 943193 N , must exceed W = 334477 N or W2 = 834029 N
Path 3-3 = 943193 N , must exceed W = 334477 N or W3 = 834029 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 9.5661 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 192.5381 mm

Input Echo, WRC297 Item 1, Description: N36" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1700.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	146.329	mm
Nozzle Thickness	54.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	2.61	MPa
Radial Load	P	-4000.00	N
Circumferential Shear	Vc	6000.00	N
Longitudinal Shear	Vl	6000.00	N
Circumferential Moment	Mc	0.270E+07	N-mm
Longitudinal Moment	Ml	0.360E+07	N-mm
Torsional Moment	Mt	0.450E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.710) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 254.329 + 2 \cdot 1.65 \cdot \sqrt{859.75(18.0 - 1.5)}$$

$$= 647.374 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1719.500 mm
Nozzle Outside Diameter	(d) =	254.329 mm
Vessel Thickness used	(T) =	16.500 mm
Nozzle Thickness used	(t) =	52.500 mm
T / t	=	0.314
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.510$$

Nr/P	=	0.083
Mr/P	=	0.175
M0/P	=	0.051
N0/P	=	0.101
MrD/Mc	=	0.298
NrDL/Mc	=	0.088
M0d/Mc	=	0.092
N0DL/Mc	=	0.063
MrD/M1	=	0.142
NrDL/M1	=	0.089
M0D/M1	=	0.045
N0DL/M1	=	0.119

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	133	136	133	136
Outplane Stress Summary	119	140	159	125

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-44	44	44	-44
Inplane Pressure Stress	66	66	66	66
Inplane Stress Summary	34	92	132	12

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2



Shear Stress Summary 3 3 1 1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	119	141	159	125

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-21	21	21	-21
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	66	66	66	66

Outplane Stress Summary 48 82 96 44

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-69	69	69	-69
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	133	136	133	136

Inplane Stress Summary 76 188 223 55

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary 1 1 3 3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	76	188	223	56

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	6	3	6



Hoop Stress Summary -1 1 10 14

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	1	1	1	1

Axial Stress Summary -2 1 8 0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary 1 1 0 0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress 3 2 11 14

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses in the
 hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	6	3	6

Hoop Stress Summary 2 5 7 10

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary -4 4 10 -3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary 0 0 1 1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	7	5	11	13

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	133	136	133	136	133	136	133	136
Circ. Pl (SUS)	-5	-5	7	7	-2	-2	4	4
Circ. Q (SUS)	-9	9	17	-17	-54	54	84	-84
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	-3	-3	5	5	-1	-1	3	3
Long. Q (SUS)	-29	29	59	-59	-17	17	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	133.0	136.0	133.0	136.0	133.0	136.0	133.0	136.0
Pm+Pl (SUS)	128.0	131.0	140.0	143.0	131.0	134.0	137.0	140.0
Pm+Pl+Q (Total)	119.0	140.1	157.1	126.0	77.1	188.0	221.0	56.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+Pl (SUS)	143.00	207.00	Passed
Pm+Pl+Q (TOTAL)	221.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	6	3	6	3	6	3	6
Circ. Pl (SUS)	-5	-5	7	7	-1	-1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-2	-2	2	2	-1	-1	1	1
Long. Q (SUS)	-2	2	4	-4	-4	4	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	3.0	6.0	3.0	6.0	3.0	6.0	3.0	6.0
Pm+Pl (SUS)	2.0	2.0	10.0	13.0	2.0	5.0	6.0	9.0

Pm+Pl+Q (Total)	3.0	1.0	10.0	14.0	6.0	5.0	8.0	13.0
-----------------	-----	-----	------	------	-----	-----	-----	------

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	138.00	Passed
Pm+Pl (SUS)	13.00	207.00	Passed
Pm+Pl+Q (TOTAL)	14.00	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K78"

From: 40

Pressure for Reinforcement Calculations	P	2.6002	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7000.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-350 LF2 +IT	
-----------------	--	----------------	--

Hub Height of Integral Nozzle	h	40.0000	mm
Height of Beveled Transition	L'	43.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		54.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	332.2720	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K78"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		7.625	in.
Actual Thickness Used in Calculation		0.500	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)



$$= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 16.2274 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.6 \cdot 98.3 / (138 \cdot 1 - 0.6 \cdot 2.6)$$

$$= 1.8741 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.8356 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 41.3 / 0.5773$$

$$= 71.5337 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	393.3500	mm
Parallel to Vessel Wall, opening length	d	196.6750	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3191.529	1051.445	NA
Area in Shell	A1	53.609	1142.247	NA
Area in Nozzle Wall	A2	769.390	855.062	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	3405.752	3405.752	NA
TOTAL AREA AVAILABLE	Atot	4328.750	5503.061	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (197 \cdot 16.2 \cdot 1 + 2 \cdot 52.5 \cdot 16.2 \cdot 1 \cdot (1 - 1))$$

$$= 3191.529 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:



$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 197(1 \cdot 16.5 - 1 \cdot 16.2) - 2 \cdot 52.5$$

$$(1 \cdot 16.5 - 1 \cdot 16.2) \cdot (1 - 1)$$

$$= 53.609 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 41.2)(11.2 - 1.87)1$$

$$= 769.390 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(41.2, 332, 40)) \cdot (54 - 12.7) \cdot 1$$

$$= 3405.752 \text{ mm}^2 \text{ Includes Bevel Area of } 101.752 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3741 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.7274 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.7274 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.1385 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 8.6600 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.66, \max(17.7, 3)]$$

$$= 8.6600 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.37, 8.66)$$

$$= 8.6600 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 32.2,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 312.8 MPa	Passed
Occasional	: 10.8,	Allowable	: 183.5 MPa	Passed
Shear	: 15.0,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 18, tr = 16.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C

Flange MDMT with Temp reduction per UCS-66(i)(2) -77 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = 2.60/5.10 = 0.510

Weld Size Calculations, Description: K78"

Intermediate Calc. for nozzle/shell Welds Tmin 16.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (3192 - 53.6 + 2 • 52.5 • 1 •
(1 • 16.5 - 16.2))138)
= 436937.50 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (4175 + 100 - 0 • 1) • 138
= 589908.69 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (4175 + 0 + 100 + (1732)) • 138
= 828969.00 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (4175 + 0 + 100 + 0 + (1732)) • 138
= 828969.00 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 302 • 10 • 0.49 • 138
= 320398. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 125) • (54 - 1.5) • 0.7 • 138
= 1984796. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 302 • (18 - 1.5) • 0.74 • 138
= 798379. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (320398 + 1984796) = 2305194 N



Failure Path [Path 2-2]:
 = (Sonw + Tpgw + Tngw + Sinw)
 = (320398 + 0 + 798379 + 0) = 1118777 N

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (320398 + 798379 + 0) = 1118777 N

Summary of Failure Path Calculations:

Path 1-1 = 2305194 N , must exceed W = 436937 N or W1 = 589908 N
 Path 2-2 = 1118777 N , must exceed W = 436937 N or W2 = 828969 N
 Path 3-3 = 1118777 N , must exceed W = 436937 N or W3 = 828969 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 13.4906 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 363.7625 mm

Input Echo, WRC297 Item 1, Description: K78" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1700.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	193.675	mm
Nozzle Thickness	54.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	2.60	MPa
Radial Load	P	-5400.01	N
Circumferential Shear	Vc	8000.00	N
Longitudinal Shear	Vl	8000.00	N
Circumferential Moment	Mc	0.480E+07	N-mm
Longitudinal Moment	Ml	0.640E+07	N-mm
Torsional Moment	Mt	0.800E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.587) is < 10
 check the Limitations of WRC bulletin 297.



Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{\text{Rmean}(t - ca)}$$

$$= 301.675 + 2 \cdot 1.65 \cdot \sqrt{859.75(18.0 - 1.5)}$$

$$= 694.720 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1719.500 mm
Nozzle Outside Diameter	(d) =	301.675 mm
Vessel Thickness used	(T) =	16.500 mm
Nozzle Thickness used	(t) =	52.500 mm
T / t	=	0.314
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.791$$

Nr/P	=	0.074
Mr/P	=	0.156
M0/P	=	0.045
N0/P	=	0.092
MrD/Mc	=	0.291
NrDL/Mc	=	0.092
M0d/Mc	=	0.090
N0DL/Mc	=	0.070
MrD/M1	=	0.122
NrDL/M1	=	0.083
M0D/M1	=	0.039
N0DL/M1	=	0.115

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	132	135	132	135
Outplane Stress Summary	113	141	167	122

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	66	66	66	66
Inplane Stress Summary	22	100	150	-1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3



Shear Stress Summary 4 4 2 2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	113	141	167	124

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-31	31	31	-31
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	66	66	66	66

Outplane Stress Summary 37 90 109 35

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-102	102	102	-102
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	132	135	132	135

Inplane Stress Summary 45 215 260 21

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary 2 2 4 4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	46	215	260	36

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7



Hoop Stress Summary -2 0 15 18

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	2	2	2	2

Axial Stress Summary -2 1 10 0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary 1 1 0 0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress 4 3 15 18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses in the
 hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7

Hoop Stress Summary 2 5 10 13

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-8	8	8	-8
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary -6 7 14 -5

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary 0 0 1 1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	9	7	14	19

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	132	135	132	135	132	135	132	135
Circ. Pl (SUS)	-7	-7	9	9	-4	-4	6	6
Circ. Q (SUS)	-13	13	23	-23	-84	84	120	-120
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	-5	-5	7	7	-3	-3	5	5
Long. Q (SUS)	-39	39	75	-75	-26	26	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	132.0	135.0	132.0	135.0	132.0	135.0	132.0	135.0
Pm+Pl (SUS)	125.0	128.0	141.0	144.0	128.0	131.0	138.0	141.0
Pm+Pl+Q (Total)	112.2	141.4	164.2	123.1	44.5	215.0	258.1	36.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	135.00	138.00	Passed
Pm+Pl (SUS)	144.01	207.00	Passed
Pm+Pl+Q (TOTAL)	258.11	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	7	4	7	4	7	4	7
Circ. Pl (SUS)	-7	-7	9	9	-3	-3	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-2	2	4	-4	-7	7	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	4.0	7.0	4.0	7.0	4.0	7.0	4.0	7.0
Pm+Pl (SUS)	3.0	0.0	13.0	16.0	1.0	4.0	9.0	12.0



Pm+Pl+Q (Total)	3.6	2.8	13.2	16.1	8.2	7.3	13.2	17.1
-----------------	-----	-----	------	------	-----	-----	------	------

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	138.00	Passed
Pm+Pl (SUS)	16.00	207.00	Passed
Pm+Pl+Q (TOTAL)	17.12	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: Boot

From: 40

Pressure for Reinforcement Calculations	P	2.6002	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	1700.00	mm
Design Length of Section	L	8233.3330	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7630.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		12.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Hub Height of Integral Nozzle	h	40.0000	mm
Height of Beveled Transition	L'	58.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		70.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Boot

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		12.000	in.
Actual Thickness Used in Calculation		0.472	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.6 \cdot 852 / (138 \cdot 1 - 0.6 \cdot 2.6)$
 $= 16.2274$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]



$$\begin{aligned}
 &= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= 2.6 \cdot 154 / (138 \cdot 1 - 0.6 \cdot 2.6) \\
 &= 2.9329 \text{ mm}
 \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.2555 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

$$\begin{aligned}
 &\text{Height value from sketch (e-1) [te]:} \\
 &= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30) \\
 &= 58 / 0.5773 \\
 &= 100.4590 \text{ mm}
 \end{aligned}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	615.6000	mm
Parallel to Vessel Wall, opening length	d	307.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Weld Strength Reduction Factor [fr1]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138 / 138) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr2]:

$$\begin{aligned}
 &= \min(1, S_n / S_v) \\
 &= \min(1, 138 / 138) \\
 &= 1.000
 \end{aligned}$$

Weld Strength Reduction Factor [fr3]:

$$\begin{aligned}
 &= \min(fr2, fr4) \\
 &= \min(1, 1) \\
 &= 1.000
 \end{aligned}$$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4994.801	1645.531	NA
Area in Shell	A1	83.899	1787.638	NA
Area in Nozzle Wall	A2	624.282	762.667	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	4783.442	4783.442	NA
TOTAL AREA AVAILABLE	Atot	5591.623	7433.747	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (308 \cdot 16.2 \cdot 1 + 2 \cdot 68.5 \cdot 16.2 \cdot 1 \cdot (1 - 1)) \\
 &= 4994.801 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 308 (1 \cdot 16.5 - 1 \cdot 16.2) - 2 \cdot 68.5 \\
 &\quad (1 \cdot 16.5 - 1 \cdot 16.2) \cdot (1 - 1) \\
 &= 83.899 \text{ mm}^2
 \end{aligned}$$



Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 41.2) (10.5 - 2.93) 1$$

$$= 624.282 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2}$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(41.2, 500, 40)) \cdot (70 - 12) \cdot 1$$

$$= 4783.442 \text{ mm}^2 \text{ Includes Bevel Area of } 143.442 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.4329 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 17.7274 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 17.7274 \text{ mm}$
Wall Thickness, shell/head, external pressure	$tr_{b2} = 2.1385 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 9.8312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[9.83, \max(17.7, 3)]$$

$$= 9.8312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.43, 9.83)$$

$$= 9.8312 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 18$, $t_r = 16.2$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = 1°C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-11 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: Boot

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 16.5000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) S_v)$$

$$= \max(0, (4995 - 83.9 + 2 \cdot 68.5 \cdot 1 \cdot (1 \cdot 16.5 - 16.2)) 138)$$

$$= 682787.50 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.



Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (5408 + 100 - 0 \cdot 1) \cdot 138 \\ &= 759987.56 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (5408 + 0 + 100 + (2261)) \cdot 138 \\ &= 1071904.50 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (5408 + 0 + 100 + 0 + (2261)) \cdot 138 \\ &= 1071904.50 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.14/2.0) \cdot 445 \cdot 10 \cdot 0.49 \cdot 138 \\ &= 472406. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.14 \cdot 188) \cdot (70 - 1.5) \cdot 0.7 \cdot 138 \\ &= 3910903. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.14/2.0) \cdot 445 \cdot (18 - 1.5) \cdot 0.74 \cdot 138 \\ &= 1177158. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (472406 + 3910902) = 4383308 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (472406 + 0 + 1177157 + 0) = 1649563 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (472406 + 1177157 + 0) = 1649563 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 4383308 N	, must exceed W = 682787 N	or W1 = 759987 N
Path 2-2 = 1649563 N	, must exceed W = 682787 N	or W2 = 1071904 N
Path 3-3 = 1649563 N	, must exceed W = 682787 N	or W3 = 1071904 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.63 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 29.6109 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 547.6108 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: MH224"

From: 50

Pressure for Reinforcement Calculations	P	2.6054	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	1700.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	18.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-350 LF2 +IT	
-----------------	--	----------------	--

Hub Height of Integral Nozzle	h	60.0000	mm
Height of Beveled Transition	L'	88.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	118.3760	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH224"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		24.000	in.
Actual Thickness Used in Calculation		0.472	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]



$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 2.61 \cdot 1703 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.61)$$

$$= 14.4718 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 2.61 \cdot 306 / (138 \cdot 1 - 0.6 \cdot 2.61)$$

$$= 5.8492 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.0036 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 88 / 0.5773$$

$$= 152.4205 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1225.2000	mm
Parallel to Vessel Wall, opening length	d	612.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.2500	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, Sn/Sv)$$

$$= \min(1, 138/138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, Sn/Sv)$$

$$= \min(1, 138/138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8865.415	1376.778	NA
Area in Shell	A1	1242.486	7354.345	NA
Area in Nozzle Wall	A2	383.688	783.456	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	7260.000	7260.000	NA
TOTAL AREA AVAILABLE	Atot	8986.174	15497.801	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1-fr1)) \text{ UG-37(c)}$$

$$= (613 \cdot 14.5 \cdot 1 + 2 \cdot 98.5 \cdot 14.5 \cdot 1 \cdot (1-1))$$

$$= 8865.415 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1



Area Available in Shell [A1]:

$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 613(1 \cdot 16.5 - 1 \cdot 14.5) - 2 \cdot 98.5 \\ &\quad (1 \cdot 16.5 - 1 \cdot 14.5) \cdot (1 - 1) \\ &= 1242.486 \text{ mm}^2 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot tlnp)(tn - trn)fr2 \\ &= (2 \cdot 41.2)(10.5 - 5.85)1 \\ &= 383.688 \text{ mm}^2 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\ &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\ &= 100.000 \text{ mm}^2 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\ &= (2 \cdot \min(41.2, 118, 60)) \cdot (100 - 12) \cdot 1 \\ &= 7260.000 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $ta = 7.3492 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= ta \\ &= 7.3492 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-46 \text{ }^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $tg = 18$, $tr = 14.5$, $c = 1.5 \text{ mm}$, $E* = 1$

Thickness Ratio $= tr \cdot E*/(tg - c) = 0.88$, Temp. Reduction $= 7 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	$-11 \text{ }^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	$-17 \text{ }^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	$-29 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-29 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	$-46 \text{ }^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(i)(2)	$-77 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

Design Pressure/Ambient Rating $= 2.61/5.10 = 0.511$

Weld Size Calculations, Description: MH224"

Intermediate Calc. for nozzle/shell Welds $Tmin \quad 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot Wo \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$$



$$= \max(0, (8865 - 1242 + 2 \cdot 98.5 \cdot 1 \cdot (1 \cdot 16.5 - 14.5)) \cdot 138) \\ = 1106989.12 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot \text{Sv} \\ = (7644 + 100 - 0 \cdot 1) \cdot 138 \\ = 1068518.75 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr}1)) \cdot \text{Sv} \\ = (7644 + 0 + 100 + (3251)) \cdot 138 \\ = 1517041.50 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot \text{tn} \cdot \text{t} \cdot \text{fr}1)) \cdot \text{S} \\ = (7644 + 0 + 100 + 0 + (3251)) \cdot 138 \\ = 1517041.50 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot \text{Wo} \cdot 0.49 \cdot \text{Snw} \\ = (3.14/2.0) \cdot 810 \cdot 10 \cdot 0.49 \cdot 138 \\ = 859846. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (\text{Dlr} + \text{Dlo}) / 4) \cdot (\text{Thk} - \text{Can}) \cdot 0.7 \cdot \text{Sn} \\ = (3.14 \cdot 356) \cdot (100 - 1.5) \cdot 0.7 \cdot 138 \\ = 10627207. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot \text{Dlo} \cdot (\text{T} - \text{cas}) \cdot 0.74 \cdot \text{Sng} \\ = (3.14/2.0) \cdot 810 \cdot (18 - 1.5) \cdot 0.74 \cdot 138 \\ = 2142596. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (859846 + 10627207) = 11487053 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (859846 + 0 + 2142596 + 0) = 3002442 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (859846 + 2142596 + 0) = 3002442 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 11487053N	, must exceed W = 1106989 N	or W1 = 1068518 N
Path 2-2 = 3002442 N	, must exceed W = 1106989 N	or W2 = 1517041 N
Path 3-3 = 3002442 N	, must exceed W = 1106989 N	or W3 = 1517041 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.62 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 53.9946 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 190.3705 mm



Input, Nozzle Desc: N22"

From: 60

Pressure for Reinforcement Calculations	P	2.6002	MPa
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	80	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	300.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-33.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	241.4260	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.6 \cdot 303 \cdot 0.89 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.6)$
 $= 2.5499$ mm



Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \left(\exp\left[\frac{P}{(Sn \cdot E)}\right] - 1 \right)$ per Appendix 1-2 (a)(1)
= $26.9 \left(\exp\left[\frac{2.6}{(138 \cdot 1)}\right] - 1 \right)$
= 0.5117 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4206 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0117	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 4.3228	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.3228	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[6.3, max(4.32, 3)]
= 4.3228 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(2.01, 4.32)
= 4.3228 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
(see 319.4.4 and 302.3.5):**

Sustained	:	11.6,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	333.4 MPa	Passed
Occasional	:	1.8,	Allowable	:	183.5 MPa	Passed
Shear	:	9.4,	Allowable	:	96.6 MPa	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
the expansion stress allowable.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govern. thk, tg = 12, tr = 2.55, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.24$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2.60/10.20 = 0.255$



Weld Size Calculations, Description: N22"

Intermediate Calc. for nozzle/shell Welds Tmin 14.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.1975 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 256.6235 mm

Input Echo, WRC107/537 Item 1, Description: N22" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	540.000	mm
Vessel Thickness	Tv	12.000	mm
Design Temperature	T1	80.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	50.800	mm
Nozzle Thickness	Tn	16.764	mm
Nozzle Material		SA-350 LF2	
Nozzle UNS Number		K03011	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Design Internal Pressure	Dp	2.600	MPa
Include Pressure Thrust		No	

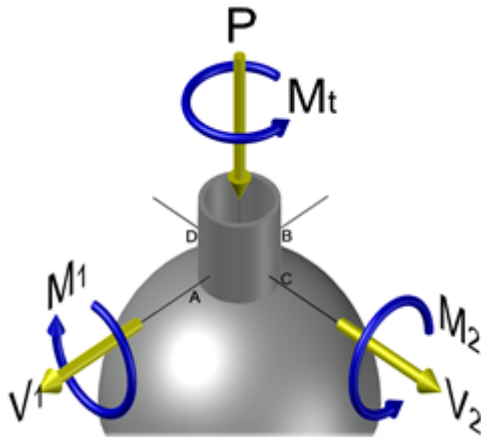
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-1300.0	N
Longitudinal Shear	(SUS)	(V1) V1	2000.0	N
Circumferential Shear	(SUS)	(Vc) V2	2000.0	N
Circumferential Moment	(SUS)	(Mc) M1	400000.0	N-mm
Longitudinal Moment	(SUS)	(Ml) M2	300000.0	N-mm
Torsional Moment	(SUS)	Mt	500000.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(276.75 (12.0 - 1.5))}$$

$$= 262.218 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-1300.0	N
Circumferential Shear	(VC) V2	2000.0	N
Longitudinal Shear	(VL) V1	2000.0	N
Circumferential Moment	(MC) M1	400000.0	N-mm
Longitudinal Moment	(ML) M2	300000.0	N-mm
Torsional Moment	MT	500000.0	N-mm

Dimensionless Param: U = 0.78 TAU = 5.00 (2.26) RHO = 0.69

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.05089	(A,B,C,D)
$M(x) / P$	SP 1	0.05557	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.07229	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.11391	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.07229	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.11391	(A,B,C,D)
$N(y) * T / P$	SP 1	0.09130	(A,B,C,D)
$M(y) / P$	SP 1	0.01795	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.08661	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.04359	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.08661	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.04359	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Rad. Bend.	P	3.9	-3.9	3.9	-3.9	3.9	-3.9	3.9	-3.9
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-4.9	-4.9	4.9	4.9
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-46.0	46.0	46.0	-46.0
Rad. Memb.	ML	-3.6	-3.6	3.6	3.6	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-34.5	34.5	34.5	-34.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-33.6	27.5	42.7	-34.2	-46.3	37.8	55.4	-44.5

Tang. Memb. P	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Tang. Bend. P	1.3	-1.3	1.3	-1.3	1.3	-1.3	1.3	-1.3
Tang. Memb. MC	0.0	0.0	0.0	0.0	-5.8	-5.8	5.8	5.8
Tang. Bend. MC	0.0	0.0	0.0	0.0	-17.6	17.6	17.6	-17.6
Tang. Memb. ML	-4.4	-4.4	4.4	4.4	0.0	0.0	0.0	0.0
Tang. Bend. ML	-13.2	13.2	13.2	-13.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-15.2	8.6	19.9	-9.0	-21.1	11.6	25.8	-12.0
Shear VC	1.4	1.4	-1.4	-1.4	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.4	-1.4	1.4	1.4
Shear MT	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3
Tot. Shear	5.7	5.7	2.8	2.8	2.8	2.8	5.7	5.7
Str. Int.	35.2	29.1	43.0	34.5	46.6	38.1	56.4	45.4

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	33.6	33.6	33.6	33.6	33.6	33.6	33.6	33.6
Rad.	Pl (SUS)	-3.0	-3.0	4.2	4.2	-4.3	-4.3	5.5	5.5
Rad.	Q (SUS)	-30.6	30.6	38.4	-38.4	-42.1	42.1	49.9	-49.9
Long.	Pm (SUS)	33.6	33.6	33.6	33.6	33.6	33.6	33.6	33.6
Long.	Pl (SUS)	-3.3	-3.3	5.4	5.4	-4.8	-4.8	6.9	6.9
Long.	Q (SUS)	-11.9	11.9	14.5	-14.5	-16.3	16.3	18.9	-18.9
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.4	1.4	-1.4	-1.4	-1.4	-1.4	1.4	1.4
Shear	Q (SUS)	4.3	4.3	4.3	4.3	4.3	4.3	4.3	4.3
Pm (SUS)		33.6	33.6	33.6	33.6	33.6	33.6	33.6	33.6
Pm+Pl (SUS)		31.9	31.9	40.1	40.1	30.6	30.6	41.4	41.4
Pm+Pl+Q (Total)		21.6	62.7	76.7	25.8	25.9	71.7	90.1	34.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.65	138.00	Passed
Pm+Pl (SUS)	41.44	207.00	Passed
Pm+Pl+Q (TOTAL)	90.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.